Torsional analysis – Level 3 – Transient forced response analysis

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Index

1. Purpose & Scope 3

2. Definitions 3

3. General procedure: description 4

4. Methodology, analysis and interpretation of results 5

5. Acceptance criteria 7

6. References 9

Annex A. Input data requirements 10

Annex B. Transient phenomena 12

Annex C. Transient torsional rotordynamic analysis with MADYN 2000 18

C.1. Madyn 2000, Transient Response Analysis 18

C.2. Fatigue Analysis, Rainflow counting cycles and Palmgren-Miner rule 33

C.3. Coupling, gears, and other components verification 42

Annex D. Report structure 44

Annex E. Case study 45

E.1. Project 24I003 – Internal Order 86988 45

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1. Purpose & Scope

This procedure is related to the dynamic analysis requirements for rotordynamic pumps, especially to the torsional analysis.

The procedure for determining the required level of detail in dynamic analysis, in alignment with best practices for pump design, is explained in **G.10.00.09** and the parameters are outlined in **Mod.401** and *ANSI/HI 9.6.8-2014*: "*if the calculated RUN is higher than 50 and less than or equal to 160, a* ***Level 3 – Advanced method – Torsional rotordynamic analysis – Transient frequency response analysis*** *is required*".

This procedure describes how to perform transient forced response (torsional) analysis of driver-coupling pump shaft train. Typical drivers are asynchronous electrical motors.

This procedure is applicable to both VS and OH or BB pumps. It is still valid for trains operated through gears and fluid couplings.

This procedure is applicable to the whole Organization.

2. Definitions

BB	Between Bearings (API610)
D, d	diameter
DOF	Degree of Freedom
E	elastic modulus
F, f	force or frequency
FEA	Finite Element Analysis
G	shear modulus
GUI	Graphical User Interface
J	moment of inertia
L	length
M	torque
n	number of cycles (fatigue verification)

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N	life in cycles
OH	Overhung (API610)
R	risk number (ANSI HI 9.6.8-2014)
RUN	R multiplied by U (ANSI HI 9.6.8-2014)
t	time
T	period
U	uncertainty value (ANSI HI 9.6.8-2014)
VFD	Variable Frequency Driver
VS	Vertically Suspended (API610)
Φ	phase
ω	pulsation
Ω	rotational speed
Goodman diagram	it predicts the fatigue life at a given stress ratio (alternating and mean stresses), based on the material properties (ultimate strength and endurance strength).
Station	boundary points of a discrete element.

3. General procedure: description

A torsional analysis must be performed on the complete shaft's train (pump, driver, coupling, gears, etc.). Torsional vibration is a fluctuating angular motion that causes twisting in couplings and shafting systems of rotating machinery.

The torsional analysis simulates how the pump, motor, coupling, and gear (if included) operate dynamically in twist excitation when run together as a system and identifies torsional natural frequencies.

Identified problems or resonances are resolved by changing the mass or inertia of components in the system (impellers, couplings, flywheels). Another alternative is to change

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the torsional stiffness of low-stiffness components, for example, by changing shaft diameters or changing the coupling type in the specification.

High torsional vibration may not be detected by typical field vibration analysis methods. High-order field vibration analysis methods, such as specially instrumented shaft contact devices or strain gauges attached to radio telemetry, are required to troubleshoot a torsional problem. Excessive or unusual coupling or bearing wear, and/or high levels of gear noise are indicators of a torsional problem. Excessive torsional vibration can lead, or contribute, to the following types of problems:

- Damaged couplings or other coupling problems.
- Gear wear, noise or tooth failure.
- Shaft fatigue or failure.
- Damage at shaft attachments, such as keys or fits.
- Loosening of motor windings.

The analysis is performed at transient conditions, such as start-up, shut-down, or other specified non-steady-state operating condition.

4. Methodology, analysis and interpretation of results

Methodology Software/Tools: • General purpose FEA programs or specialty rotordynamic programs. The tool described in this procedure is: • Madyn 2000
Model See ***SOP-460*** *"Level 2 Undamped torsional natural frequency analysis"*. The analysis should include damping (primarily from the pump impeller oscillating and dissipating energy based on experience). A modal damping of 1% is a typical value.
Analysis Mode shapes, angular displacements, and stresses are calculated at critical points along the drive train. All forcing functions should be applied using the worst-case scenario of location and direction.

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Interpretation of results

See ***SOP-460*** *"Level 2 Undamped torsional natural frequency analysis"*.

In addition, an assessment of angular displacements and stresses are calculated at critical locations along the drive train.

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5. Acceptance criteria

As indicated in API 610, an appropriate cumulative fatigue algorithm shall be used to develop a value for the safe number of starts and/or short circuit events. Values between 1000 to 1500 starts are common. API 541 requires 5000 starts, which is a reasonable assumption for a motor. The driven equipment, however, would be overdesigned to meet this requirement.

The transient response analysis should generate dynamic torsional torques and stresses to evaluate the mechanical integrity of the components (mainly shafts and couplings). The maximum torques shall be used to evaluate the peak torque capability of coupling components, gearings and interference fits of components, such as coupling hubs, and shaft locations with stress raisers (API 610). *Figure 1* summarizes the torsional analysis procedure.

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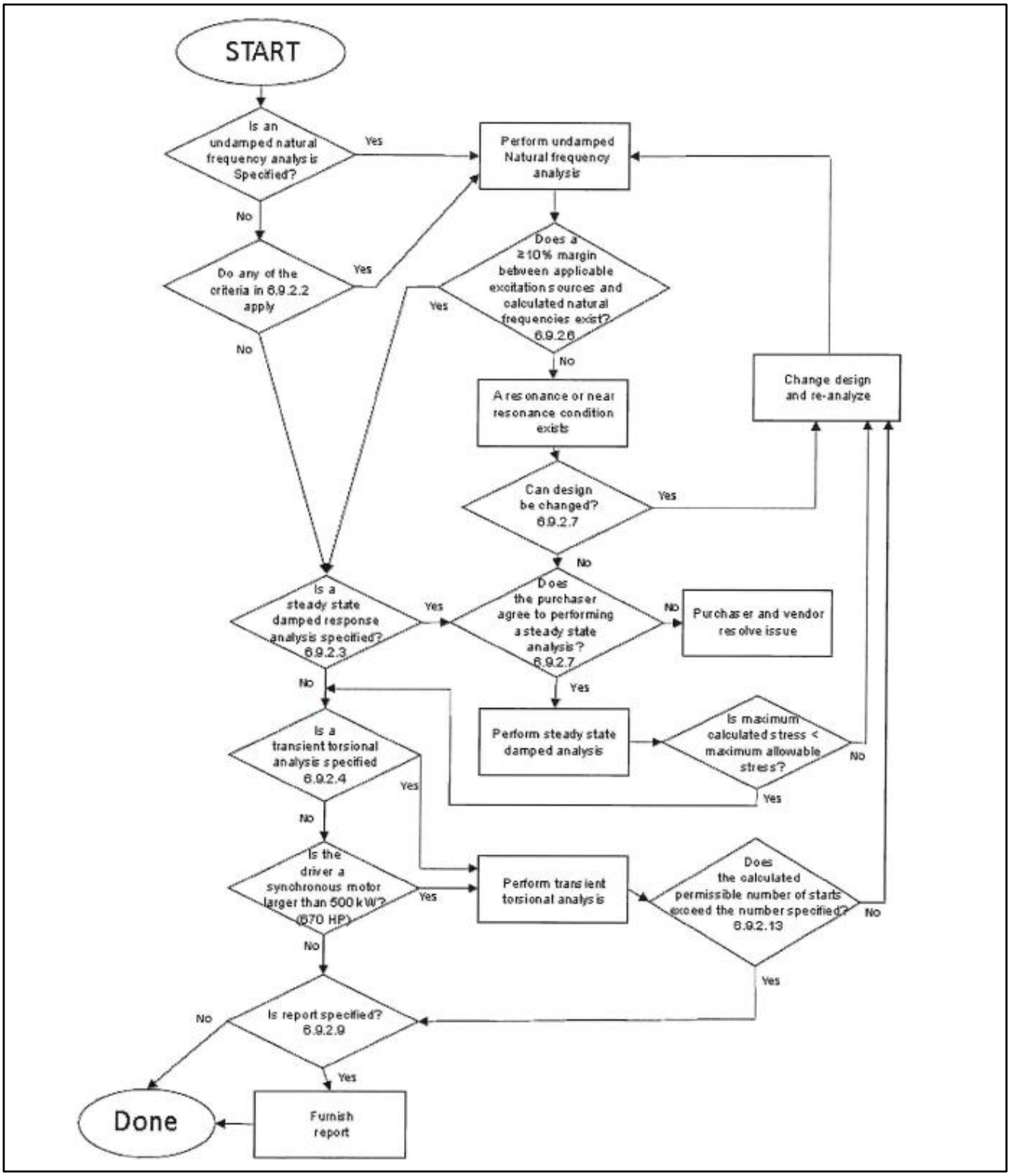

Figure 1. Torsional Analysis Flow Chart

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6. References

API Standard 610, 12th Edition, January 2021

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ANSI/HI 9.6.8-2014 Rotordynamic Pumps – Guideline for Dynamic Pumping Machinery

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Corbo, Mark A., and Stanley B. Malanoski. "Practical Design Against Torsional Vibration." Proceedings of the 25th Turbomachinery Symposium. Texas A&M University. Turbomachinery Laboratories, 1996.

G.10.00.09 – Guidelines for dynamic analysis complexity level

SOP-460 – Level 2 undamped torsional natural frequency analysis

Mod.401 – Spreadsheet linked to procedure G.10.00.09 (in agreement with ANSI HI 9.6.8)

Mod.539 – Transient torsional

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Annex A. Input data requirements

The following information is typically required for a torsional rotordynamic analysis:

1. Pump's shaft dimensions (drawing preferred)
2. Rotor's material of construction:
   - Shaft
   - Impellers
   - Sleeved
   - Coupling
3. Mass and inertia (polar) for all rotating components:
   - Impellers
   - Sleeves
   - Mechanical seal rotating components

Note: mass and inertia can be estimated from component geometry if necessary.

4. Assembly information:
   - Axial location of all rotating components on the shaft
   - Diametral fit of all rotating components on the shaft (loose or press, and if press, how much diametral interference is present)
   - Length and location of all keys
5. Coupling information (drawing preferred, and usually available from vendor):
   - Manufacturer, type and size
   - Coupling torsional stiffness (as a function of speed if applicable)
   - Details of spacer (length, mass and inertia)
   - Coupling hub fit-up length and degree of press fit, length of key
6. Motor information:
   - Rotor drawing
   - Rotor mass and inertia (polar)
   - Rotor core fit-up on shaft (loose or press, and if press, how much diametral interference is present)
   - Motor fan mass and inertia (polar)
   - Motor fan fit-up on shaft

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- Length and location of key
- Torsional stiffness of the motor stub shaft between the motor core end and coupling hub

7. Diesel information:

- Polar inertia of all rotating and reciprocating components (i.e., flywheel, coupling, pistons, etc.)
- Torsional stiffness of the shaft
- Number of power strokes per revolution
- Fit of coupling's shaft (clearance or press fit)
- Running speed

8. Turbine information:

- Rotor drawing or sketch showing key shaft dimensions and envelope dimensions of each turbine wheel
- Rotor's materials of construction – shaft, turbine wheels, sleeves
- Mass and inertia (polar) of rotation components
- Number of blades on turbine wheel

9. Right-angle gear information:

- Gear drawings
- Shaft torsional stiffnesses
- Gear polar inertias
- Gear ratio
- Number of gear teeth

General information needed for transient response analysis:

1. Transient information

- Airgap torque at start-up, three-phase short circuit, line-to-line phase short circuit, phase opposition reclosing and other transient conditions if required by the customer.

2. Driving and counter torque curve during run up conditions as a function of speed.

3. If VFD run-up: the harmonic order of pulsating torques, and for each exciting harmonic, the amplitude and the phase as a function of speed.

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4. If asynchronous run-up: the amplitude and the phase of pulsating torque as a function of speed.
5. Coupling max peak torque.

Annex B. Transient phenomena

During the **run-up** of asynchronous motors, the torque varies according to speed. The motor torque, also called driving torque, should win the load torque, or counter torque, to allow the rotor to accelerate.

Initially, at standstill, the motor produces high starting torque, which is necessary to overcome inertia and any load. As the motor accelerates, the torque typically increases to a peak (breakdown torque) before decreasing to the rated torque at operating speed (*Figure 2*).

The starting current is significantly higher than the running current, often 4 to 7 times the full-load current. This high inrush current can cause voltage dips. In this phase, motors are often protected by devices such as soft starters or Variable Frequency Drives (VFDs) to limit the inrush current and reduce mechanical stress (see **start-up** part).

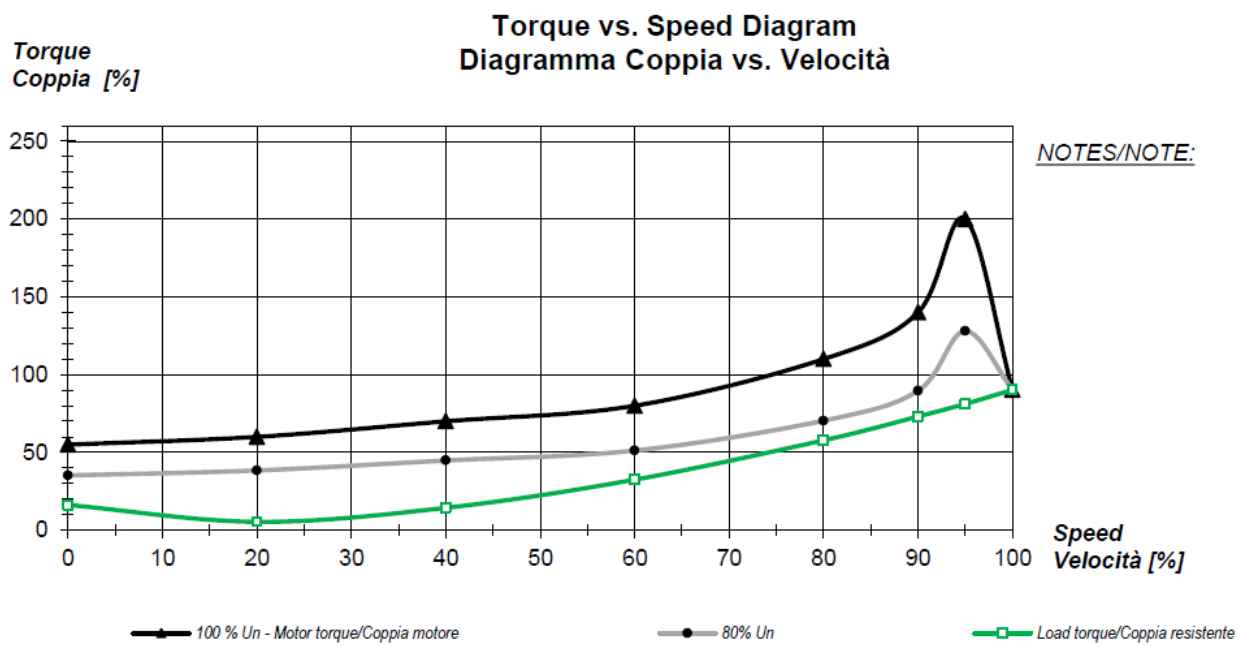

Figure 2. Asynchronous run-up torque-speed characteristic

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The driving torque can be viewed as the sum of steady-state torque M_D and a pulsating torque M_O, and both are a function of the speed (n). The pulsating torque has a pulsation Φ related to the slip frequency. It is the difference between the synchronous speed (the speed of the rotating magnetic field) and the actual rotor speed. At standstill, slip is 1 (or 100%), and it decreases as the motor speeds up.

$$M(t) = M_D\big(n(t)\big) + M_0\big(n(t)\big) \cdot \sin\Big(\phi\big(n(t)\big)\Big)$$

The counter torque $M_c\big(n(t)\big)$ is a function of the rotating speed and it is simplified without considering the fluctuating part.

Motor + VFD run-up is similar to the asynchronous motor one. The main difference is in the pulsating torque. In the asynchronous motor its frequency depends on the slip value; while for the VFD, the frequency depends on the electrical and mechanical excitations introduced by the VFD. The manufacturer VFD should provide information about these excitations both in terms of frequency, amplitude and phase.

The manufacturer VFD does not always provide the information related to the steady-state driving torque. When provided, it is often the maximum torque that the VFD can deliver during run-up, which might not always be realistic compared to an actual on-site torque, or the worst case. Therefore, the suggested procedure is to create a run-up curve based on the pump's manufacturer experience and customer's requirements. For example, it can be a simple constant torque. Based on the load and the motor curve, a certain run-up time is obtained; this is the time it takes for the motor to reach its operating speed. Adjust the load torque iteratively until the run-up time that is plausible for the application is found.

The worst run-up time case is not always obvious. For some trains, a slow run-up is the worst case because the train crosses the resonances slowly and the response can grow. Indeed, during a fast crossing of the resonances the response does not have enough time to grow. For some other trains, the worst case is when the VFD delivers its maximum possible run-up torque because there is a big torque step at the beginning of the run-up.

Start-up. For induction motors (asynchronous motors), excitation occurs at the very beginning of the start-up (when the breaker is closed), and the motor is suddenly energized (switch-on). Over a short time (a few hundreds of a second), the transient torques are very

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high. The train has not really started to accelerate yet. The counter torque can be considered null since the rotor is standstill.

The driving torque usually has a constant value and sinusoidal terms that decrease with time (*Figure 3*). The excitation frequency is equal to the line frequency (e.g. 50 Hz or 60 Hz). Usually, there is no mode with a natural frequency equal to this frequency. However, even if they have good separation margins, some modes can be excited by this sudden high-amplitude excitation (typically, for a motor-gear-load train, the 1st mode has the highest response).

The nature of the phenomenon is time-transient and, as mentioned above, the train does not really accelerate. Therefore, an excitation as a function of time is provided by the motor manufacturer. Additionally, it can be used since the cause of the high response is not a resonance.

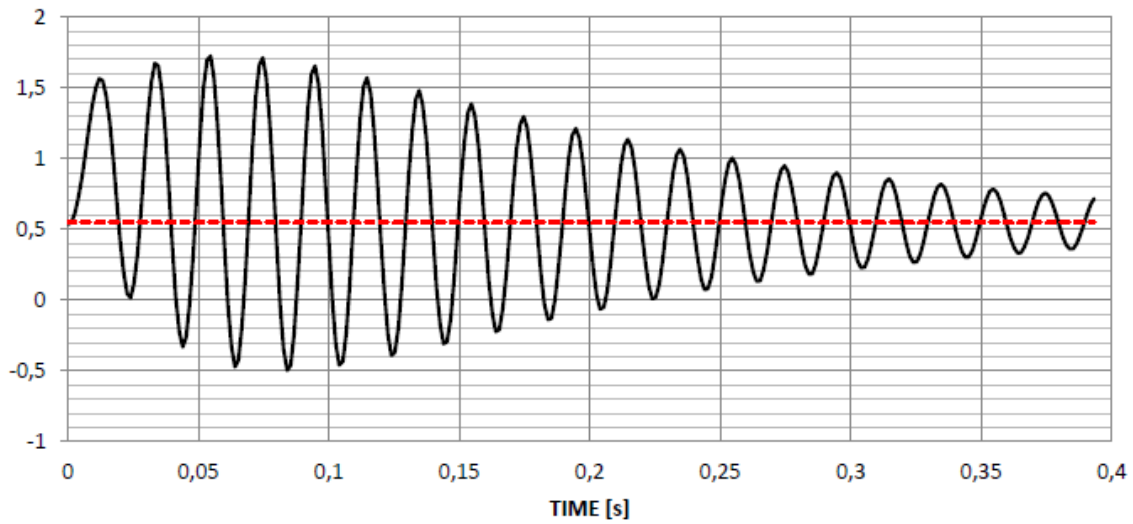

Figure 3. Transient torque at start-up

The **<u>three-phase short circuit</u>** occurs when all three phases (conductors) are being short-circuited together. It is a symmetrical fault, meaning that the fault current is balanced and equal in all three phases. The magnitude of this torque depends on various factors, including: the motor's design, the impedance of the motor windings, and the characteristics of the power supply. During a short circuit, the high current flow generates a strong magnetic field, resulting in a large torque that can cause mechanical stress and potential damage to the motor components.

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The driving torque M_D is a pulsating function decreasing with time. The excitation frequency is equal to the line frequency (*Figure 4*). The counter torque is equal to the rated torque.

$$M_D = M_1 \cdot e^{-\frac{t-t_1}{T_1}} \cdot \sin(\omega(t-t_1)+\phi_1)$$

A three-phase short circuit can occur due to several reasons, including:

- Insulation Failure: degradation of insulation between the windings can lead to short circuits. This can be caused by factors like aging, overheating, or exposure to contaminants.
- Mechanical Damage: physical damage to the motor windings or connections, often due to improper handling or maintenance, can result in short circuits.
- Overvoltage Conditions: high-voltage spikes or surges can break down the insulation and cause short circuits.
- Contamination: conductive contaminants, such as dust, moisture, or chemicals, can bridge the insulation and cause a short circuit.
- Overheating: excessive heat can degrade the insulation material, leading to short circuits.

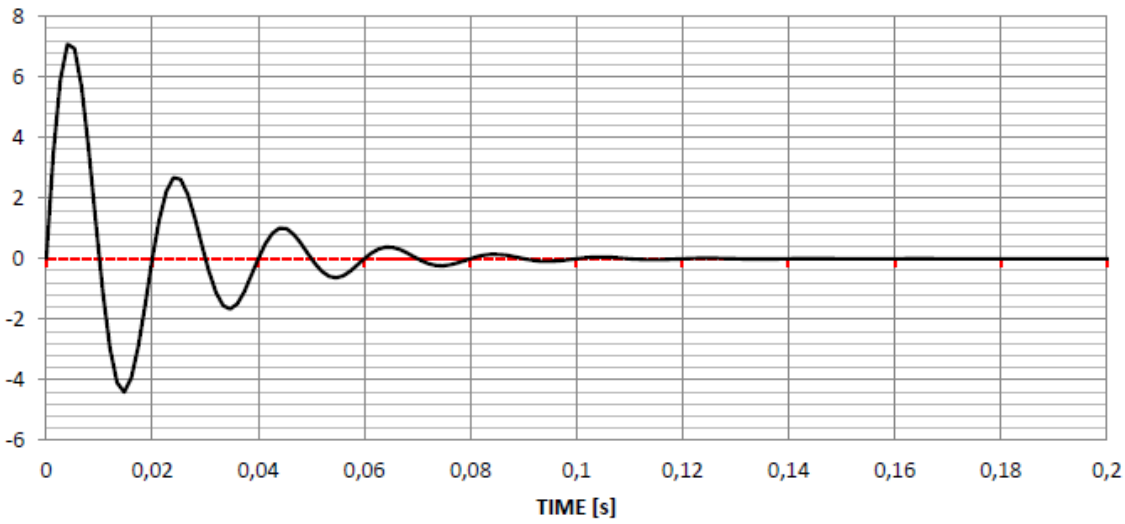

Figure 4. Transient torque at three phase short circuit

A **line-to-line short circuit** reaction torque in electric motors occurs when two of the three phases are short-circuited together. It is an asymmetrical fault, meaning that the fault current is not balanced and only affects the two involved phases.

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The driving torque M_D is a sum of pulsating functions decreasing with time. The excitation frequency is equal to the sum of the line frequency and twice the line frequency (*Figure 5*). The counter torque is equal to the rated torque.

$$M_D = M_1 \cdot e^{-\frac{t-t_1}{T_1}} \cdot \sin(\omega(t-t_1)+\phi_1) + M_2 \cdot e^{-\frac{t-t_1}{T_2}} \cdot \sin(2\omega(t-t_1)+\phi_2)$$

A line-to-line short circuit can occur due to several reasons like a three-phase short circuit.

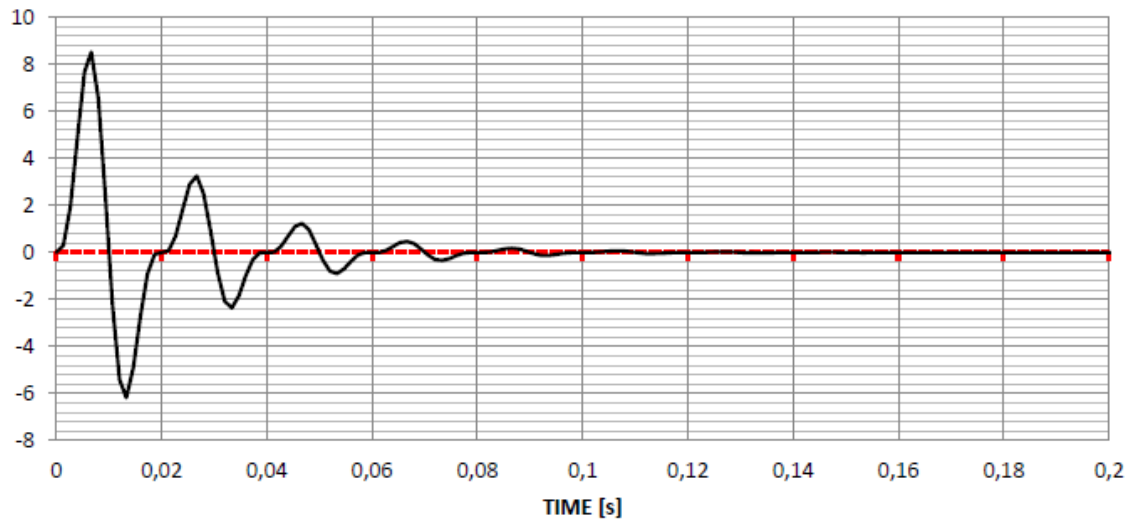

Figure 5. Transient torque at line-to-line short circuit

Three-phase short circuits, result generally in the highest fault currents and are considered the most severe type of fault. Whereas Line-to-line short circuits, result typically in lower fault currents compared to a three-phase short circuit. The fault current is usually about 87% of the three-phase fault current.

Three-phase short circuit is less common but more severe. It can occur due to major insulation failures or catastrophic events. Line-to-line short circuit is more common and can occur due to factors like heavy winds causing conductors to touch, or insulation failures between two phases.

These events can happen during both start-up and steady-state operations, but they are more likely to occur during steady-state when the motor is running under load.

A **phase opposition reclosing (or automatic reclosing)** reaction torque occurs when an electric motor is re-energized out-of-phase with the power source. This can happen during events like residual voltage transfers or momentary voltage interruptions. When the

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motor is reconnected to the power source with a significant phase angle difference, it can cause high transient currents and torques. The reason is that the motor's residual magnetic field and the phase difference between the motor and the power source can cause large inrush currents and torques.

The driving torque is usually the sum of a constant term, and a variable and sinusoidal terms that decrease with time. The excitation frequency for the pulsating term is equal to the line frequency.

$$M_D = M_{01} + M_{02} \cdot e^{-\frac{t-t_1}{T_0}} + M_1 \cdot e^{-\frac{t-t_1}{T_1}} \cdot \sin(\omega(t-t_1) + \phi_1)$$

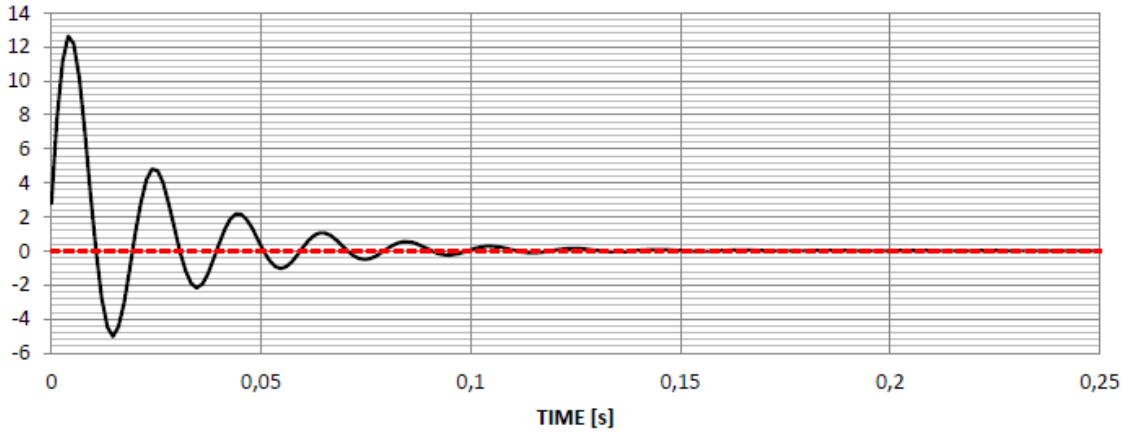

Figure 6. Transient torque at phase opposition reclosing

Phase opposition reclosing reaction torque can occur both at the start-up and during the steady state operation, but it is more commonly associated with events like momentary voltage interruptions or residual voltage transfers during steady state. At the start-up, the specific issue of phase opposition reclosing is less common because the motor is not yet synchronized with the power source.

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Annex C. Transient torsional rotordynamic analysis with MADYN 2000

C.1. Madyn 2000, Transient Response Analysis

Transient loads are a combination of force and moment components, which can be applied to stations. The main types of transient loads (*Figure 7*) used are:

- Transient Force f(t), TFT
- Transient Force f(n), TFN

"Transient Force f(t)" loads have an excitation force function f(t) describing a time history (*Table 1*). It is mainly used for fault conditions, such as short circuit.

Table 1. Excitation function of Type f(t)

Name of the function	Formula
Rectangular_Function	$f(t) = R(t)$ $R(t) = 0\ for\ t < t_1$ $R(t) = A\ for\ t \geq t_1$ $R(t) = 0\ for\ t > t_2$
Sine_Cosine_Series	$f(t) = R(t)\left(\sum_k A_{0k} \cdot e^{-\frac{t-t_1}{T_{0k}}} + \sum_i A_i \cdot e^{-\frac{t-t_1}{T_{A_i}}} \cdot \sin(\omega_i(t-t_1) + \phi_{si}) + \sum_j B_j \cdot e^{-\frac{t-t_1}{T_{B_j}}} \cdot \cos(\omega_i(t-t_1) + \phi_{si})\right)$
Short_Circuit_2Phase	$f(t) = R(t)\left(M_1 \cdot e^{-\frac{t-t_1}{T_1}} \cdot \sin(\omega(t-t_1) + \phi_1) + M_2 \cdot e^{-\frac{t-t_1}{T_2}} \cdot \sin(2\omega(t-t_1) + \phi_2)\right)$
Short_Circuit_3Phase	$f(t) = R(t)\left(M_1 \cdot e^{-\frac{t-t_1}{T_1}} \cdot \sin(\omega(t-t_1) + \phi_1)\right)$
Fault_Synchronization	$f(t) = R(t)\left(M_{01} + M_{02} \cdot e^{-\frac{t-t_1}{T_0}} + M_1 \cdot e^{-\frac{t-t_1}{T_1}} \cdot \sin(\omega(t-t_1) + \phi_1)\right)$

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Switch_On	$f(t) = R(t)\left(M_{01} + M_{02} \cdot e^{-\frac{t-t_1}{T_0}} + M_1 \cdot e^{-\frac{t-t_1}{T_1}} \cdot \sin(\omega(t-t_1) + \phi_1) + M_2 \cdot e^{-\frac{t-t_1}{T_2}} \cdot \sin(2\omega(t-t_1) + \phi_2)\right)$

"Transient Force f(n)" loads have an excitation, which is defined as a function of speed and the time history of the excitation itself, that must be calculated via the speed as a function of time n(t) (*Table 2*). This type of transient load can be used for moments around the rotor axis. The function has a driving and counter torque. The speed as a function of time is determined by integrating the rotor run-up from the moments driving the rotor, the moments representing the counter torque(s) and the total moment of inertia of the system. The driving and counter torque in the excitation function must be positive. The correct sign must be defined by the moment component(s) applied to the station(s). This load can be used to define the transient run-up.

Table 2. Excitation function of the Type f(n)

Name of the file	Formula
Asynchronous Run-up	$f(n(t)) = M_D(n) + M_c(n) + M_0(n) \cdot \sin\left(\phi(n(t))\right)$ $\phi(n(t)) = 2i_p \int (\Omega_{nom} - \Omega(t))dt$ Where: • n *is the speed* • $M_D(n)$ *is the driving torque* • $M_c(n)$ *is the counter torque* • $M_0(n)$ *is the oscillating torque* • n_{nom} *is the nominal speed* $[Hz]$ • i_p *is number of pole pairs*
VFD Run-up	$f(n(t)) = M_D(n) + M_c(n) + \sum_i M_{n_fLine_i}(n) \cdot \cos\left(\phi_i(n(t))\right)$ $\phi_i(n(t)) = \int \left(i_{fLine} \cdot \Omega_{Line}(t) - i_n \cdot \Omega(t)\right) dt + \Phi_{n_fLine_i}$ Where:

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	• n *is the speed* • f_{Line} *is the line frequency* • $M_D(n)$ *is the driving torque* • $M_c(n)$ *is the counter torque* • i_{fLine} *is a multiple of line frequency* • i_n *is a multiple of speed* • $M_{n_fLine_i}(n)$ *is the oscillating torque* • $\Phi_{n_fLine_i}$ *is the phase of oscillating torque*

A remark concerning asynchronous run-up: the reaction between nominal speed and the number of pole pairs is fixed. For this reason, only the nominal speed can be defined in the excitation GUI.

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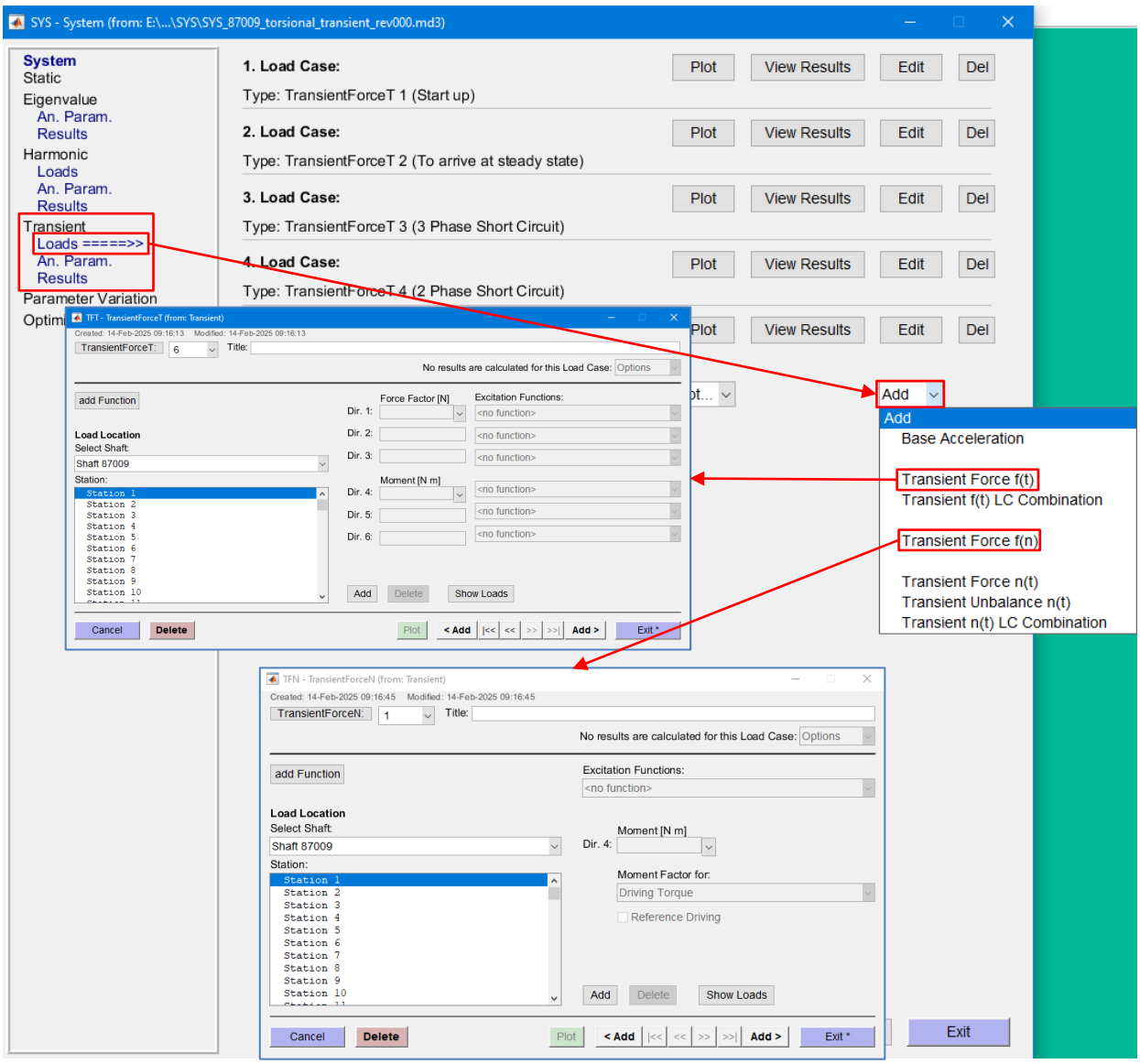

Figure 7. GUI Torsional Load

For induction motors, the excitation at **start-up** happens over a short time when the train has not really started to accelerate. This means that the counter torque on the pump is 0 and there is only a driving torque on the motor core.

As shown in *Figure 8*, it is needed to:

- Use Transient Force f(t), TFT.
- Define a function by selecting the most appropriate function type. “Sine Cosine Series” or “Switch On” functions from the list are the ones suggested for this condition.

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Insert the correct parameters. The motor supplier should provide the values or at least the equation to define them. The excitation frequency is equal to the line frequency (e.g. 50 Hz or 60 Hz).

- Select the station to apply the driving torque. It can be applied at the center of the motor's core or distributed along the motor's core.
- Add to that station the torque value, and the excitation function. They should be defined along the rotation axis (direction 4).

It is always suggested to check the defined excitation function. It should correspond to the one provided by the motor supplier. It can be verified it with the "Plot" command in the function excitation GUI.

It is also recommended to verify the correct location of the load with the "Plot" command in the transient load GUI.

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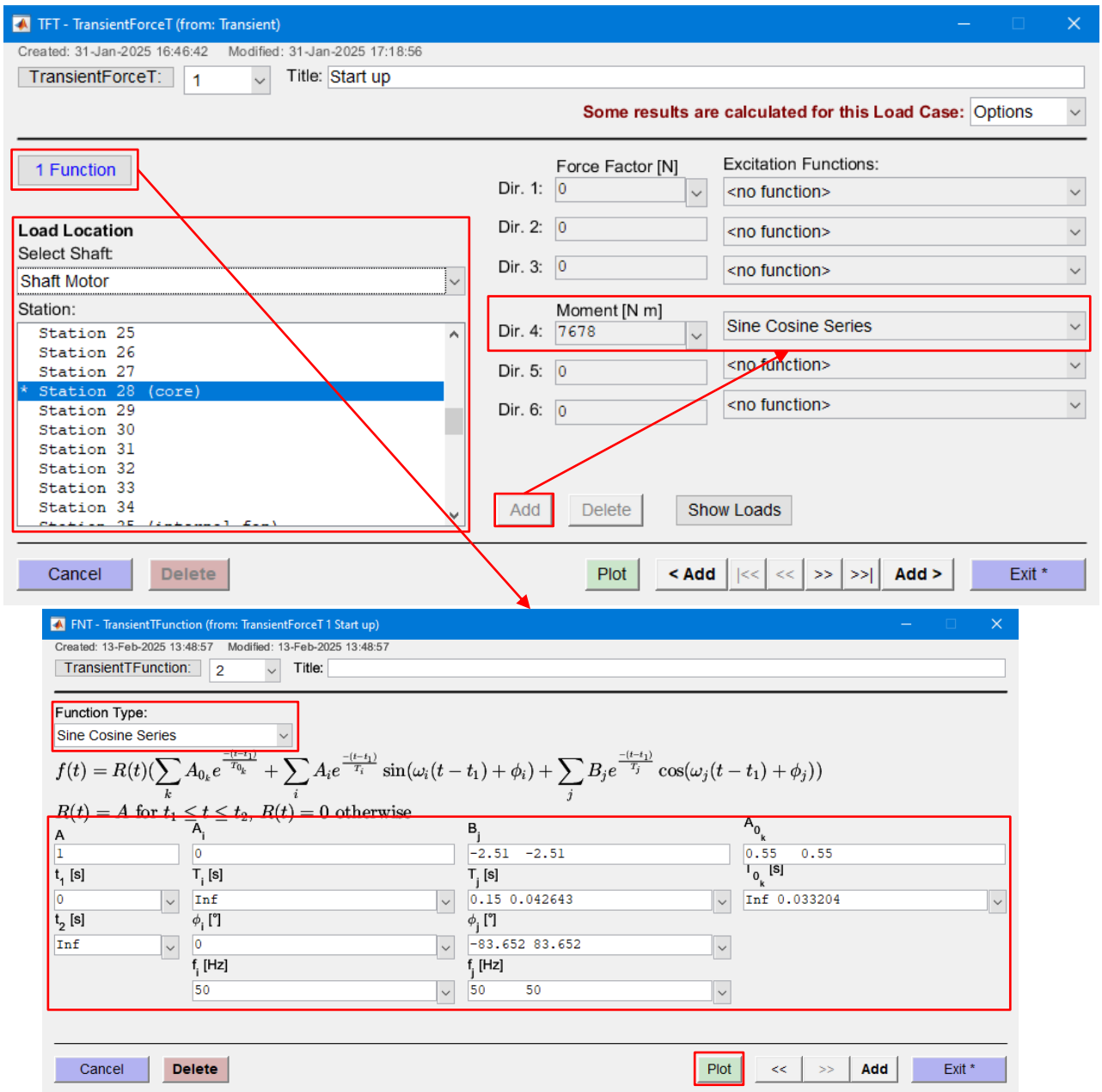

Figure 8 Star-up excitation asynchronous motor

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For **transients at steady state**, such as short circuit or phase opposition reclosing, the counter torque on the pump is not 0, but equal to the rated torque.

The steps to follow (Figure 9*)* are similar to those listed above:

- Use Transient Force f(t), TFT.
- Define a function by selecting the most appropriate function type. “Sine Cosine Series”, “Short Circuit 2 Phase”, “Short Circuit 3 Phase” and “Fault Synchronization” are the most used to define the driving torque. A “Rectangular” function is recommended for the counter torque.
  Insert the correct parameters for each case. The motor supplier should provide the values or at least the equation to define them. If transients are related to electrical phenomena, the excitation frequency is equal to the line frequency (e.g. 50 Hz or 60 Hz).
- Select the stations to apply the driving and counter torque. Apply the driving torque at the center of the motor’s core or distributed along the motor’s core. Apply the counter torque at the stations where the impellers are positioned.
- Add to those stations the torque value and the excitation function. Check that the driving and counter torque have an opposite effect. In case of driver and driven are directly coupled, torque values should have opposite sign.
  Note that the torque applied at a train must be calculated with the speed of that train and applied it along the rotation axis (direction 4).

To initialize the transients at steady state, define a **“dummy” load** using a “Rectangular” function for both driving and counter torque as shown in *Figure 10*.

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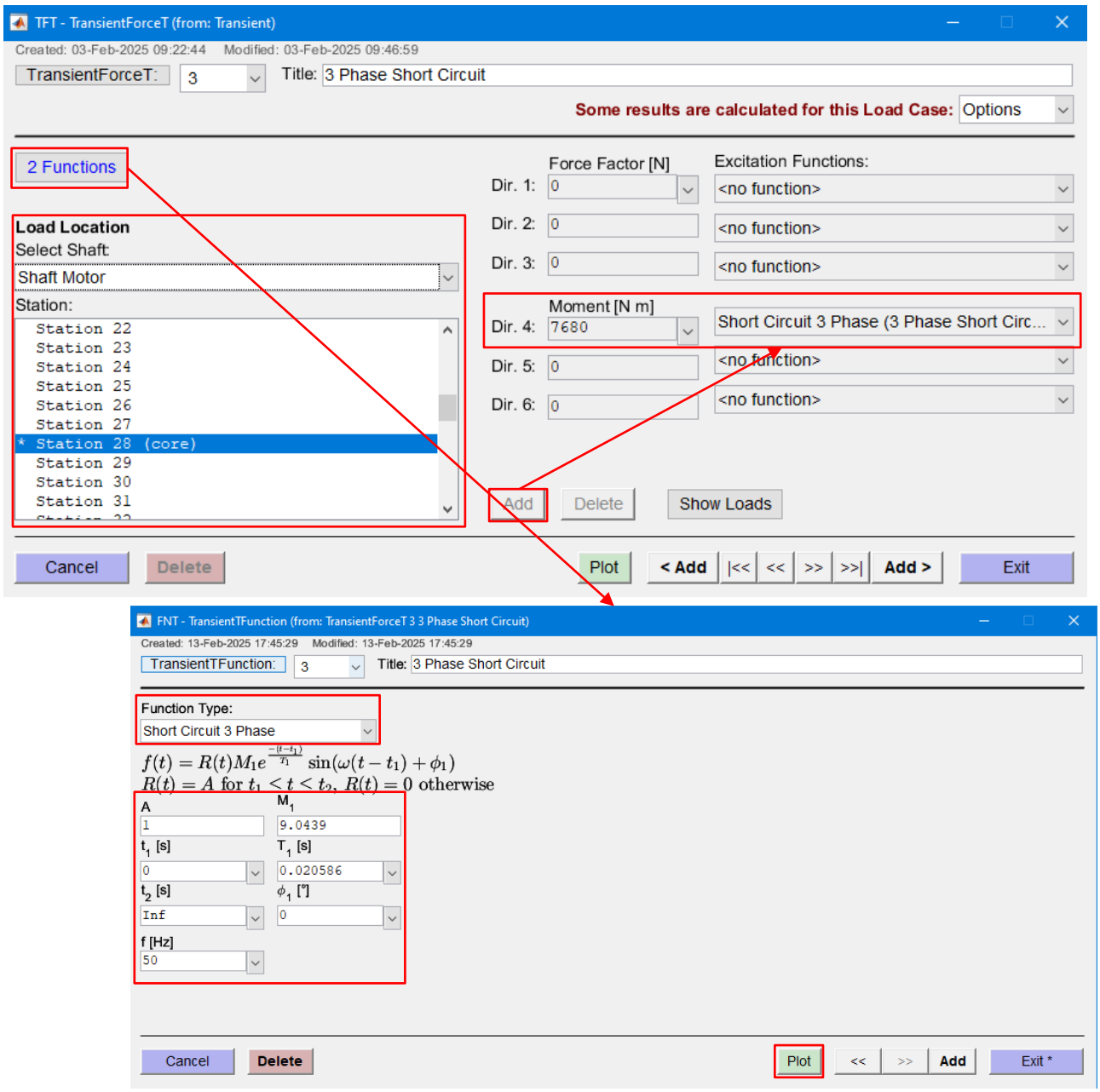

Figure 9. Driving torque excitation of transient at steady state

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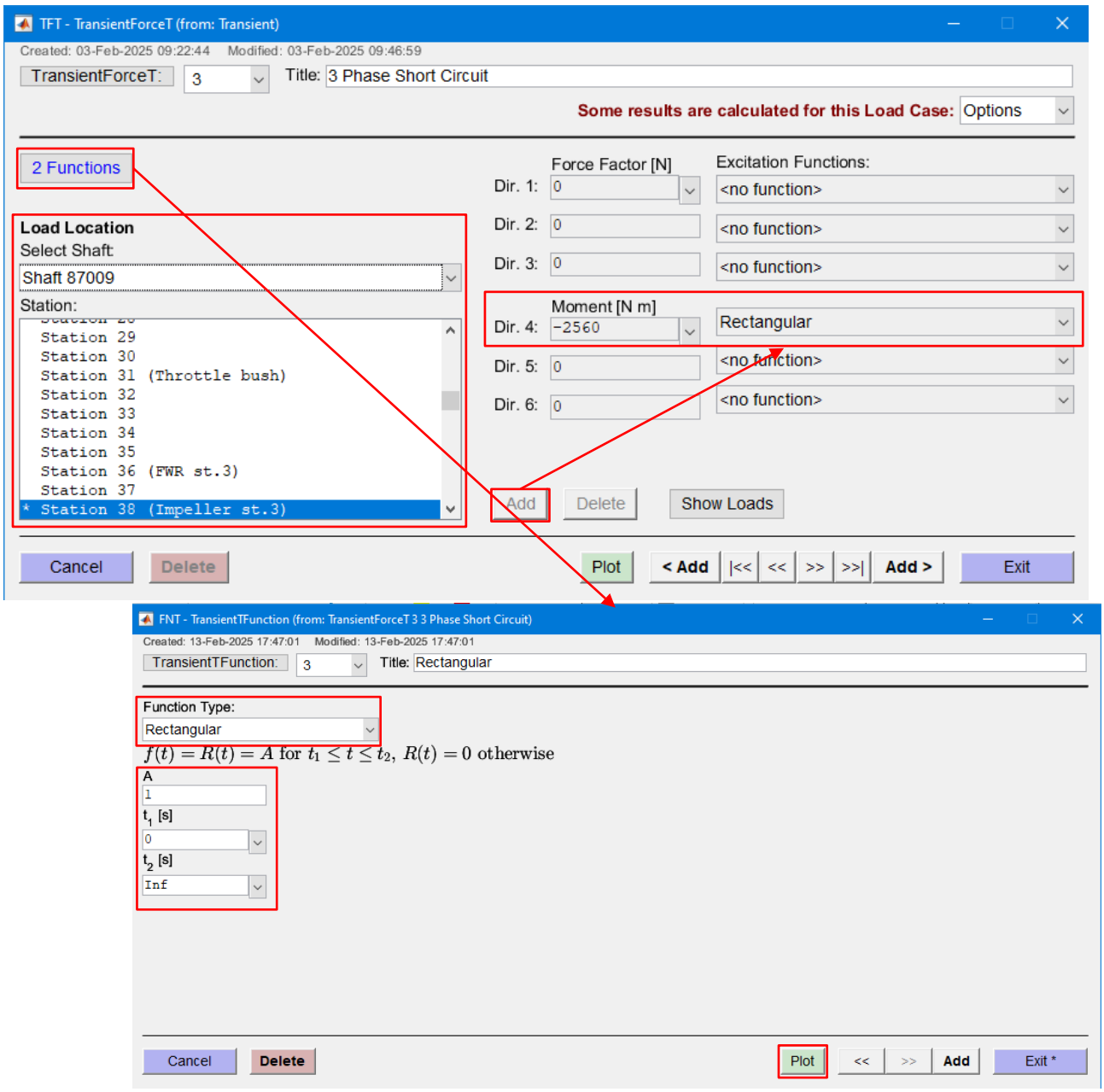

Figure 10. Counter torque excitation of transient at steady state

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Run-up excitation is defined as a function of speed. The excitation load to be used is different from the previous one:

- Use Transient Force f(n), TFN
- Define a function by selecting the most appropriate function type. Use “Asynchronous Run Up” function for induction motors without VFD and “VFD Run Up” function for induction motors with VFD.

  Insert the right parameters. Driving and counter (and oscillating torque for “Asynchronous Run Up”) torques require several inputs equal to the number of speeds defined. In “VFD Run Up”, edit the oscillating torque defining the multiple of speed and/or line frequency, the oscillating torque and phase as a function of the speed. Define a combination of excitations.
- Select the stations to apply the driving and counter torques. Apply the driving torque at the center of the motor’s core or distributed along the motor’s core, and the counter torque at the stations where the impellers are positioned.
- Add to those stations the torque value and the excitation function. Note that the torque applied at a train must be calculated with the speed of that train and it must be applied along the rotation axis (direction 4).

  For the driven train, set the moment factor as “Counter Torque”, while for the driver, set it as “Driving Torque”. Check the box “Reference Driving” on the driver station. Verify that the driving and counter torques have an opposite effect. In case of driver and driven directly coupled, torque values should have an opposite sign.

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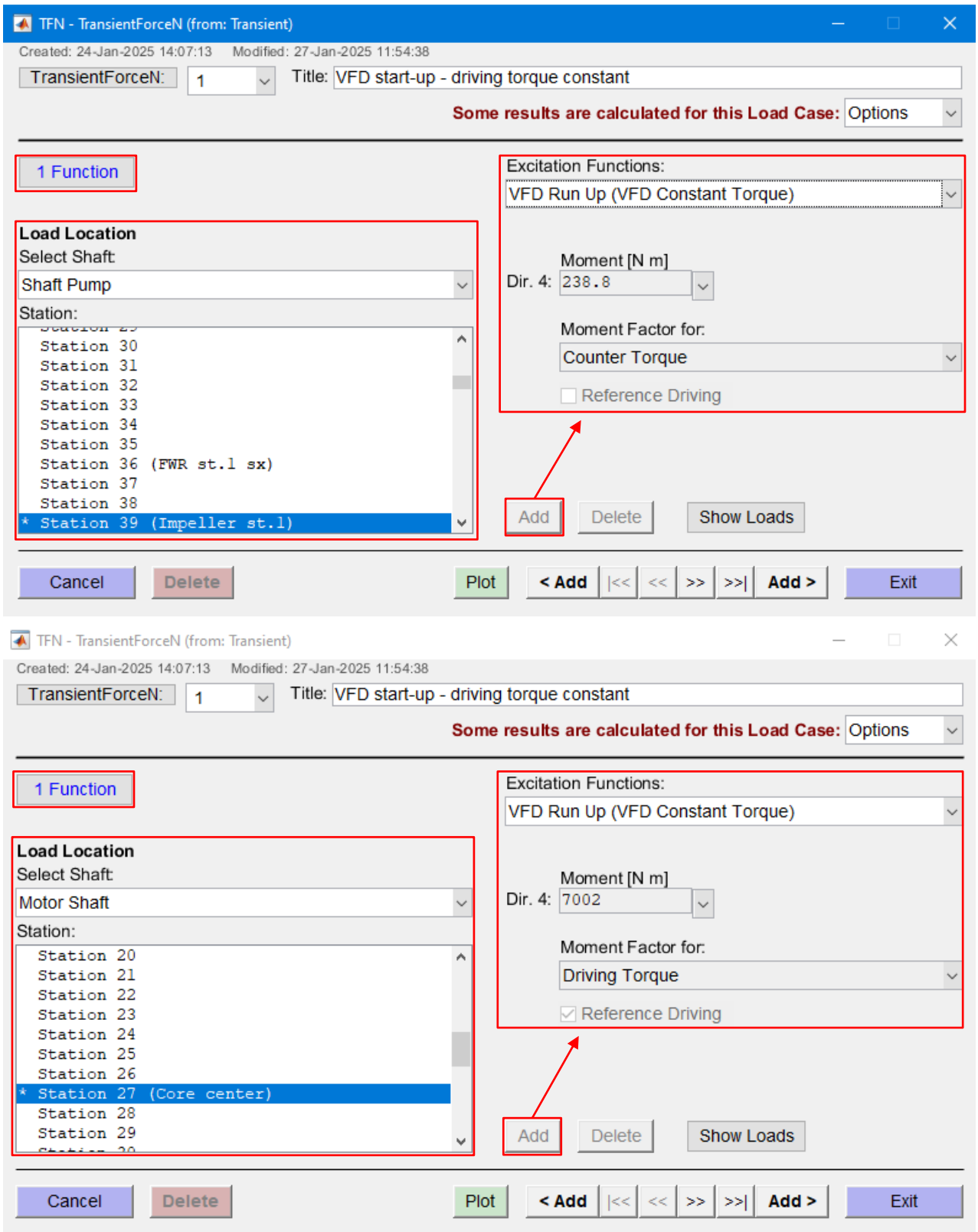

Figure 11. Counter and driving torque excitation of run up

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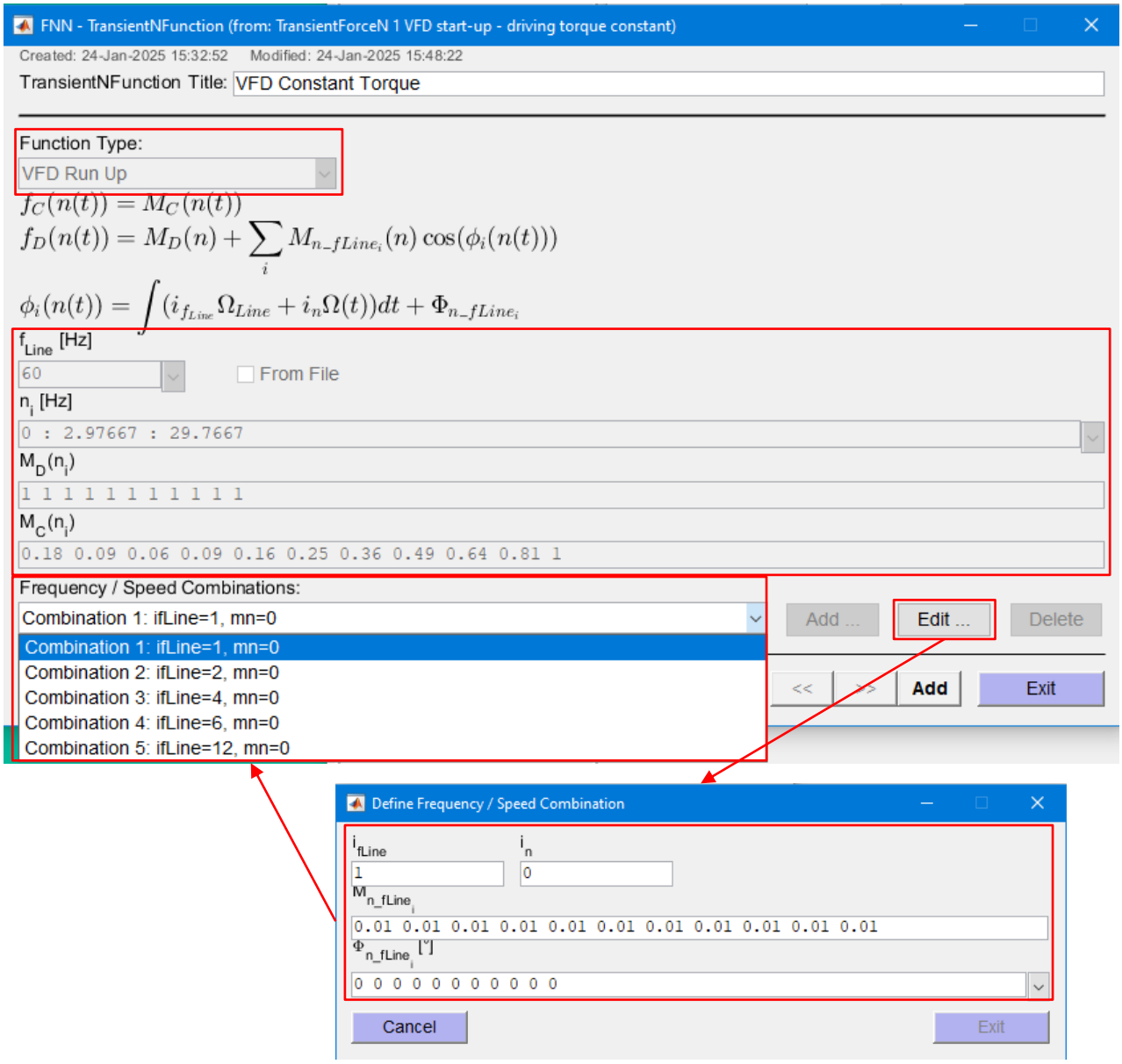

Figure 12. Run up excitation function

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The transient response is calculated by means of a bimodal reduction. This means that the natural modes from an eigenvalue analysis must exist and must be selected to carry out the transient analysis.

Additional parameters and settings can be defined in the transient response analysis GUI:

1. **Initial conditions**. Either zero initial conditions or results from another transient analysis can be selected. For initial conditions from transient analyses, the end time is taken as default.

   For start-up or run-up conditions, a zero initial condition can be selected; for fault conditions at steady state, such as short circuit, initial conditions from another transient analysis can be used. In this second case a "dummy" transient response analysis must be created to initialize the desired transient condition. In this case use a zero initial condition.
2. **EIG Results**. Select the torsional natural frequencies from an eigenvalue analysis.
3. **Excitation Type**. In the radio button menu select the right excitation type.
4. **Maximum Frequency**. Defines the limit for the consideration of modes. Inf is the default parameter to use.
5. **Modal Damping**. It defines a common value for all modes in the selected frequency range. The suggested value is 1% for all conditions. In the "dummy" scenario a higher value can be defined (10%) to quickly arrive at steady state.
6. **Load Case**. Select the transient load to simulate in the analysis.
7. **Start time, end time and time step**. For f(n) excitation only, the time step is defined. The criteria in *Table 3* define an adequate time step to apply for the different types of excitations.

   The time steps are also checked against the maximum selected natural frequency. The following criterion is applied:

   $$\Delta t = 1/(2f_{max})$$

   End time should be defined such that the system after the transient excitation returns at the steady state condition. If the steady condition has not been reached, increase the end time value.

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Table 3. Adequate time steps for different types of excitation

Excitation Function	Criterion for Time Step Δt	Remark
Rectangular Function	For $t_2 < 1s$ & $(t_2 - t_1) \cdot 400/10 < 1$: $\Delta t = (t_2 - t_1)/10$ For $t_2 < 1s$ & $(t_2 - t_1) \cdot 400/10 > 1$: $\Delta t = 1/400$ For $t_2 > 1s$: $\Delta t = 1/400$	$t_2 - t_1$ is the time span of the rectangle
Sine Cosine Series	$\Delta t = 1/(10 f_{max})$	f_{max} maximum occurring frequency
2 Phase Short Circuit	$\Delta t = 1/(20 f)$	f electric supply frequency
3 Phase Short Circuit	$\Delta t = 1/(10 f)$	f electric supply frequency
Fault Synchronization	$\Delta t = 1/(10 f)$	f electric supply frequency
Switch On	$\Delta t = 1/(20 f)$	f electric supply frequency
Asynchronous Run Up	$\Delta t = 1/(20 n_{nom})$	n_{nom} nominal motor speed
VFD Run Up	$\Delta t = 1/(2 f_{max})$	f_{max} maximum occurring frequency among line frequency f_{line}, speed n and $i_n n + i_{fLine} f_{Line}$

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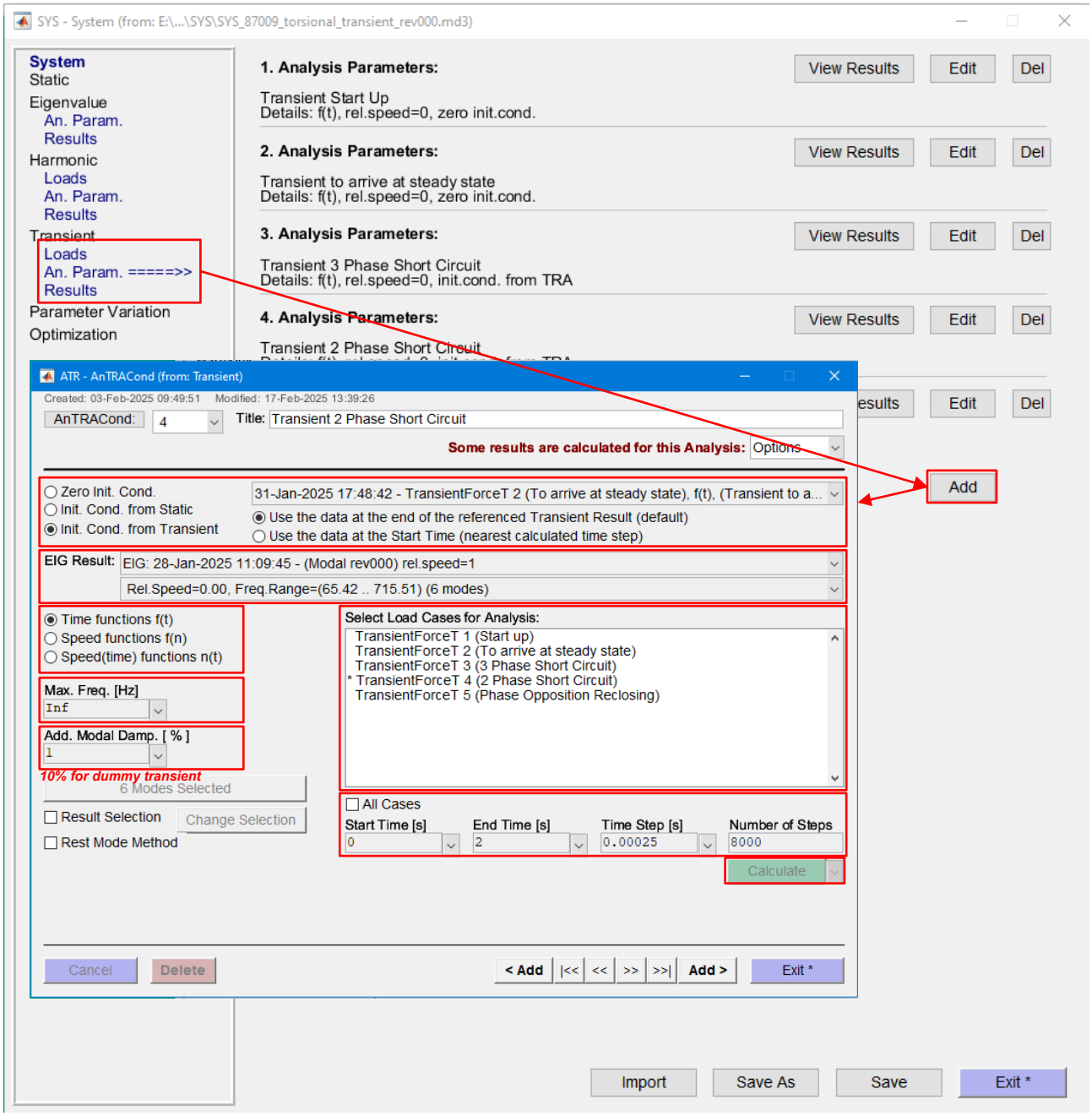

Figure 13. GUI Torsional Analysis

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C.2. Fatigue Analysis, Rainflow counting cycles and Palmgren-Miner rule

Transient loading patterns depart from the sinusoidal character of a smoothly rotating machinery. Cycle counting is used to summarize (often lengthy) irregular load-versus-time histories by providing the number of times cycles of various sizes occur. Cycle counts can be made for time histories of force, stress, strain, torque, acceleration, deflection, or other loading parameters of interest. The definition of a cycle varies with the method of cycle counting. ASTM E1049-85 provides standard practices for cycle counting methods employed in fatigue analysis. This part covers the procedure used to obtain cycle counts by rainflow counting.

Rules for the rainflow counting method are as follows: let *X* denote the range under consideration; *Y* the previous range adjacent to *X*; and *S* the starting point in the history.

1. Read next peak or valley. If out of data, go to Step 6.
2. If there are less than three points, go to Step 1. Form ranges *X* and *Y* using the three most recent peaks and valleys that have not been discarded.
3. Compare the absolute values of ranges *X* and *Y*.
   (a) If $X < Y$, go to Step 1.
   (b) If $X \geq Y$, go to Step 4.
4. If range *Y* contains the starting point *S*, go to Step 5; otherwise, count range *Y* as one cycle; discard the peak and valley of *Y*; and go to Step 2.
5. Count range *Y* as one-half cycle; discard the first point (peak or valley) in range *Y*; move the starting point to the second point in range *Y*; and go to Step 2.
6. Count each range that has not been previously counted as one-half cycle.

The load history is plotted in *Figure 14a* and it is used to illustrate the process. Details of the cycle counting are as follows:

1. S = A; Y = |A-B|; X = |B-C|; X > Y. Y contains S, that is, point A. Count |A-B| as one-half cycle and discard point A; S = B. (See *Figure 14b*)
2. Y = |B-C|; X = |C-D|; X > Y. Y contains S, that is, point B. Count |B-C| as one-half cycle and discard point B; S = C. (See *Figure 14c*)
3. Y = |C-D|; X = |D-E|; X < Y. Read next peak.
4. Y = |D-E|; X = |E-F|; X < Y. Read next peak.

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5. Y = |E-F|; X = |F-G|; X > Y. Count |E-F| as one cycle and discard points E and F. (See *Figure 14d*. Note that a cycle is formed by pairing range E-F and a portion of range F-G).
6. Y = |C-D|; X = |D-G|; X > Y; Y contains S, that is, point C. Count |C-D| as one-half cycle and discard point C. S = D. (See *Figure 14e*).
7. Y = |D-G|; X = |G-H|; X < Y. Read next peak.
8. Y = |G-H|; X = |H-I|; X < Y. End of data.
9. Count |D-G| as one-half cycle, |G-H| as one-half cycle, and |H-I| as one-half cycle. (See *Figure 14f*)
10. End of counting. See the table in *Figure 14* for a summary of the cycles counted in this example.

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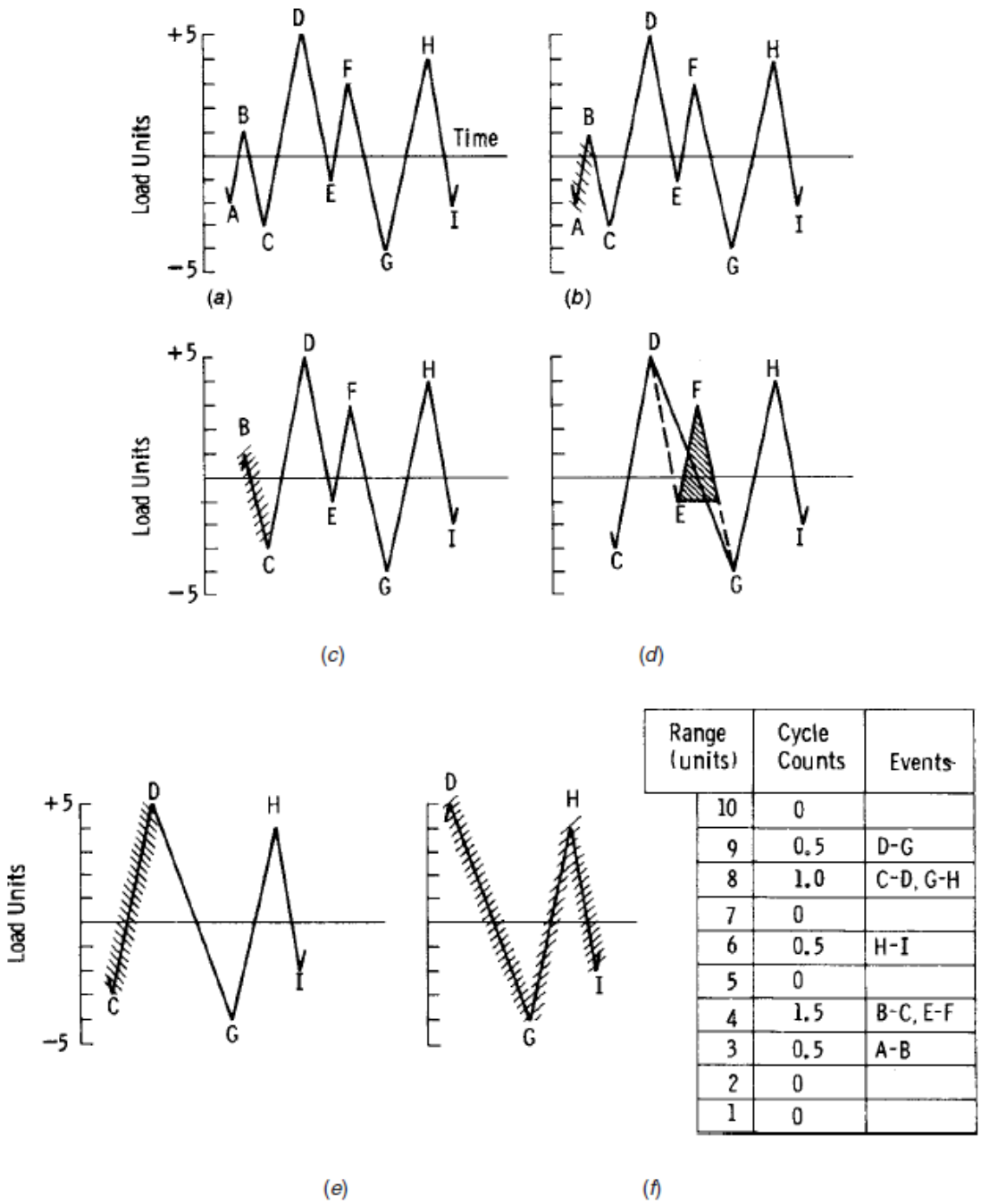

Range (units)	Cycle Counts	Events
10	0	
9	0.5	D-G
8	1.0	C-D, G-H
7	0	
6	0.5	H-I
5	0	
4	1.5	B-C, E-F
3	0.5	A-B
2	0	
1	0	

Figure 14. Rainflow counting example

Predicting the life of parts subjected to randomly varying stress cycle intensity, such as transient, is complicated. The procedure, called the *linear cumulative-damage rule* or *Palmgren-Miner rule* try to estimate this life.

The simple concept of this rule is that if a part is cyclically loaded at a stress level causing failure in 10^5 cycles, each cycle of this loading consumes one part in 10^5 of the life of the

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component. If other stress cycles are interposed corresponding to a life of 10^4 cycles, each of these consumes one part in 10^4 of the life, and so on. When 100% of the component's life has been consumed on this basis, fatigue failure is predicted.

The *Palmgren-Miner rule* is expressed by the following equation in which n_1, n_2, …, n_k represent the number of cycles at specific overstress levels, and N_1, N_2, …, N_k represent the life (in cycles) at the overstress levels, as taken from the appropriate S-N curve. Fatigue failure is predicted when:

$$\frac{n_1}{N_1} + \frac{n_2}{N_2} + \cdots + \frac{n_k}{N_k} = 1 \; or \; \sum_{j=1}^{j=k} \frac{n_j}{N_j} = 1$$

Mod.539 – *Transient torsional* combines the rainflow counting method and *Palmgren-Miner* rule to estimate the shaft's life subjected to a transient load. It is a python script that uses python module *rainflow.py* to count cycles in fatigue analysis according to ASTM E1049-85.

Mod.539 – *Transient torsional* requires different parameters as input (see *Figure 15)*. In the first section, material properties such as tensile strength and yield strength are required. Moreover, other input parameters required for the fatigue analysis, such as shaft diameter, working temperature, surface type finishes and a reliable value of endurance limit are requested.

In the second section, it is required:

- The filename of stresses during transient condition, called **inputfile.**
- The filename of fatigue stress concentration factor, called **kfile.**
- The output filename, **fileout.**
- The shaft name of which the fatigue analysis has been performed, **shaft_name**. It is the shaft name inside the **inputfile.**
- The limit for fatigue failure, called **alfa_limit**. The default value is 1.
- The shaft element for which the script will create a Goodman diagram, called **graph**.

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```
# Material
Su = 660.            # ultimate tensile stress [MPa]
Sy = 520.            # yield stress [MPa]
diameter = 140.      # shaft diameter [mm]
temp = 107.           # temperature [°C]
surf = 2             # 1 = mirror polished;
                     # 2 = fine ground or commercially polished
                     # 3 = machined or cold drawn
                     # 4 = hot-rolled
                     # 5 = as forged
reliab = 4           # 1 = 50%
                     # 2 = 90%
                     # 3 = 99%
                     # 4 = 99.9%

# Input
inputfile = "TRA_87009_torsional_transient_2phase_rev000.txt"
kfile = "k_factor.txt"
fileout = "results.txt"
shaft_name = "Shaft 87009"
alfa_limit = 1.      # limit for Miner's rule
graph = 46           # element for which create graph
```

Figure 15. **Mod.539** – *Transient torsional input parameters*

The **inputfile** of transient response analysis can be created for each shaft as shown in *Figure 16*. The file contains the **shaft_name** and the shear stresses for each shaft element (left and right stations) and for each timestep in the range considered (*Figure 17*). These torsional stresses are used to create a shear stress vs. time curve for each element and, for the selected element **graph,** a Goodman diagram. For more information about the Goodman diagram refer to **SOP-460** *"Level 2 Undamped torsional natural frequency analysis"*.

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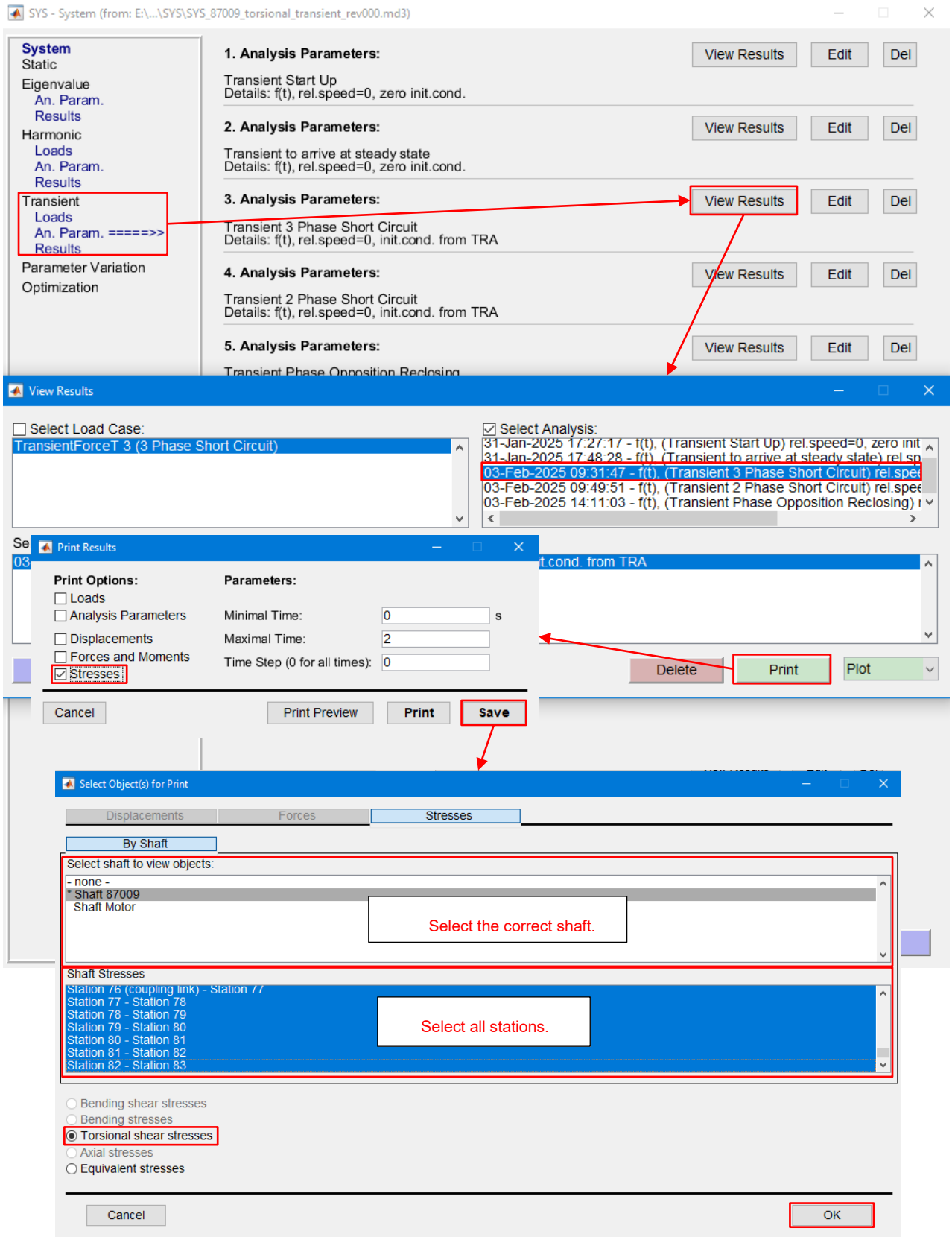

Figure 16. GUI to extract transient stress

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```
Shaft Stresses
    t [s]      t4 [N/m2]
Shaft 87009, Station 1 - Station 2
  0.000e+00   -4.167e+03
              4.598e+03  (right)
  2.500e-04   -4.334e+03
              4.455e+03  (right)
  5.000e-04   -4.497e+03
              4.253e+03  (right)
  7.500e-04   -4.653e+03
              4.001e+03  (right)
  1.000e-03   -4.801e+03
              3.782e+03  (right)
  1.250e-03   -4.940e+03
              3.666e+03  (right)
  1.500e-03   -5.073e+03
              3.621e+03  (right)
```

Figure 17. Transient stress vs time

The **kfile** contains the fatigue stress concentration factor to account for the presence of notch in the shaft. Shear stress are increased proportionally to these factors. Refer to **SOP-460** *"Level 2 Undamped torsional natural frequency analysis"* for more information about the fatigue stress concentration factor.

The input file should have on each row the element identification number and the stress concentration factors for the left and right stations of the element (*Figure 18*).

```
ele ID,k_left,k_right
1,1,1
2,2,2
3,1,1
4,1,1
5,2,2
6,1,1
7,2,2
8,3,3
9,3,3
10,2,2
11,1,1
12,2,2
13,2,2
14,1,1
15,1,1
```

Figure 18. Fatigue stress concentration factor input file

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After running **Mod.539** – *Transient torsional*, the script gives as output an image and the **fileout**.

The image contains, on the left, the shear stress curves in function of time of the left and right stations of the selected element, on the right, a Goodman diagram has been plotted for the stations of the selected element (*Figure 19*). The green region represents the infinite-life region, meaning the component can support those stress combinations for at least 10^6 cycles. The yellow region is the area corresponding to a life of 10^3 cycles.

It is recommended that the stress combinations of the most stressed element are at least inside the yellow region to guarantee a life of the component greater than 10^3 cycles.

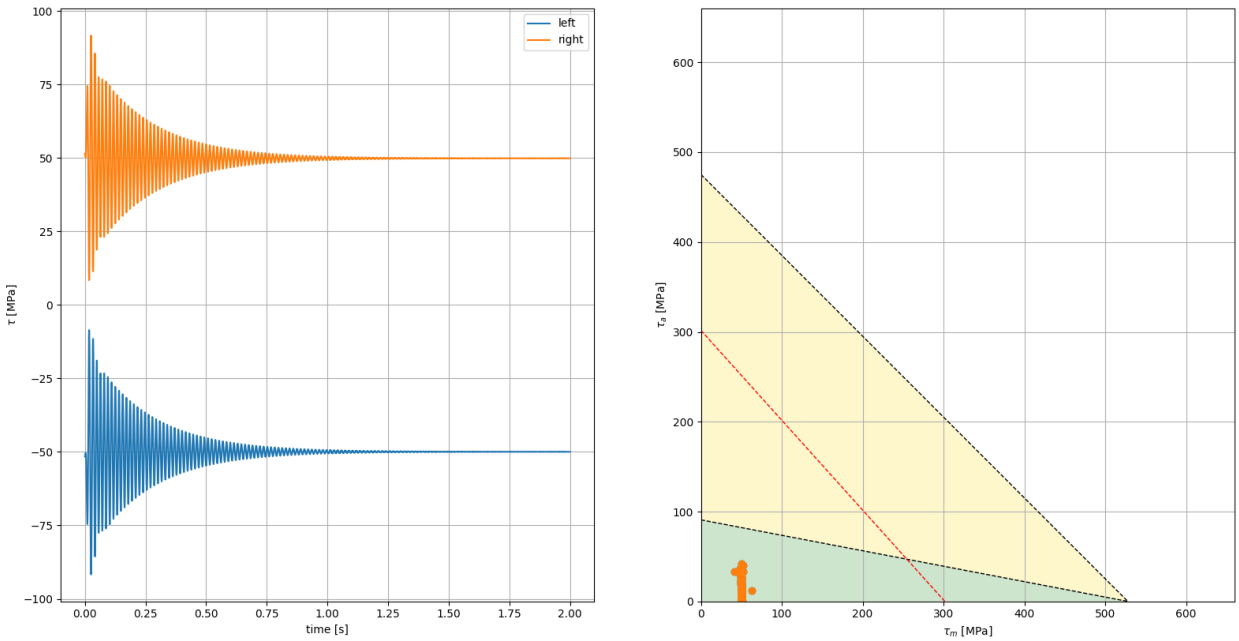

Figure 19. Shear stress curve (left). Goodman diagram (right)

The output file **fileout** contains in each row the element identification number, the identification for the left or right station, the fatigue value resulting from the *Palmgren-Miner rule* and a warning message.

Verify that for each station the warning message says “ok”. If a warning message saying “not ok” appears, a fatigue failure of the component can happen. In this case, the transient should be revised if possible. For example, a different run-up curve can be defined through a VFD run-up, or a new design of the component should be prepared as a last resource.

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```
EleID    Side  alfa        Message
1        left  0.000000    ok
1        right 0.000000    ok
2        left  0.000000    ok
2        right 0.000000    ok
3        left  0.000000    ok
3        right 0.000000    ok
4        left  0.000000    ok
4        right 0.000000    ok
5        left  0.000000    ok
5        right 0.000000    ok
6        left  0.000000    ok
6        right 0.000000    ok
7        left  0.000000    ok
7        right 0.000000    ok
8        left  0.000000    ok
8        right 0.000000    ok
```

Figure 20. Output file **Mod.539** *– Transient torsional*

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C.3. Coupling, gears, and other components verification

Another control performed is the verification of the torque occurring in components such as gears and couplings during transient conditions. The coupling max torque (M_{max}) is the absolute value reached due to a transient load (*Figure 21*). This value should be compared with the maximum peak torque (M_{peak}) provided by the supplier. It is the maximum torque the component can tolerate for short periods, such as transients. Thus, it is recommended to keep the maximum transient torque below the maximum peak torque for these components.

$$|M_{max}| < M_{peak}$$

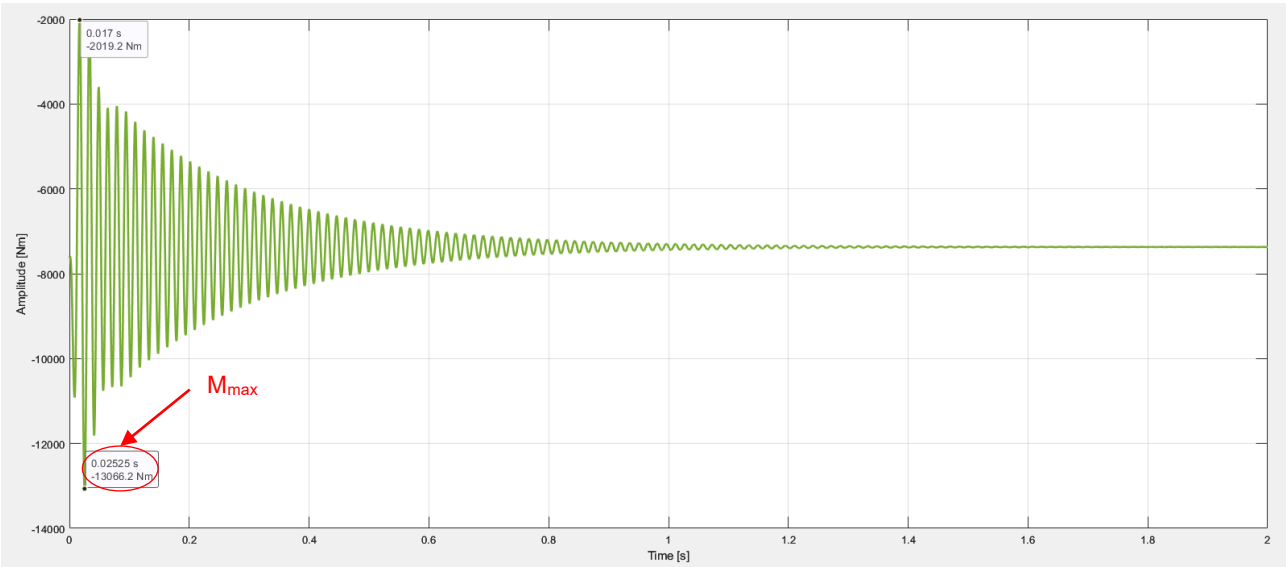

Figure 21. Coupling transient torque

Figure 22 show the process to plot the transient torque. The image contains the maximum peak torque to use for the check.

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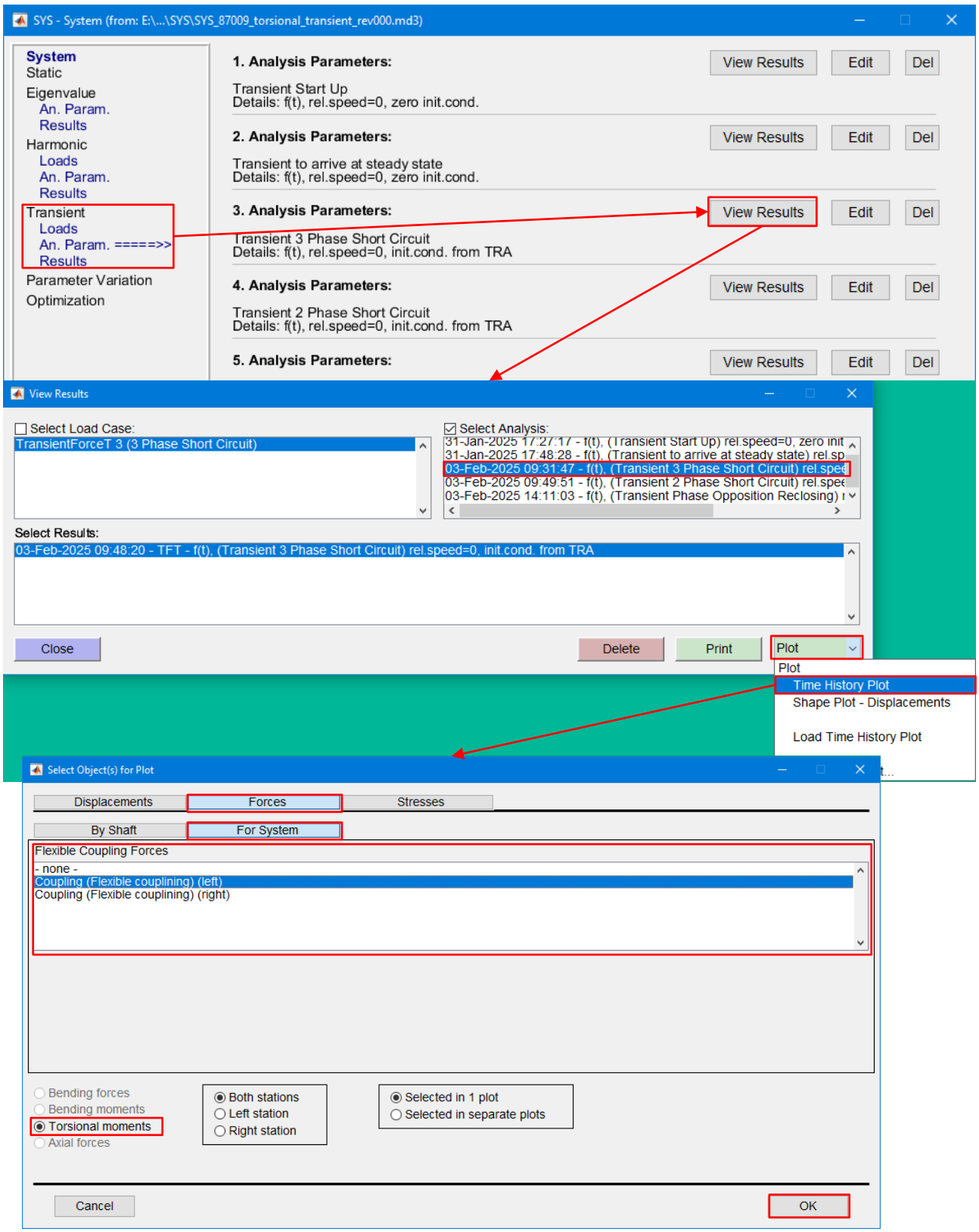

Figure 22. GUI to plot coupling transient torque

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Annex D. Report structure

Refer to **SOP-460** *"Level 2 Undamped torsional natural frequency analysis"* for the report structure. Add at the end of the **numerical analysis** part a paragraph called:

TRANSIENT ANALYSIS. It should include, for the different transient conditions analyzed, the input parameters and the equations used to define the driving and counter torque.

It should show that the coupling maximum transient torque is lower than the peak torque provided by the supplier for these components.

A fatigue verification through a Goodman diagram should demonstrate that the pump shaft has the alternate and mean stress of transient inside the region of 10^3 cycle. It should also state that, according to *Palmgren-Miner rule,* a fatigue life of at least 10^3 cycles for the transient to be respected.

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Annex E. Case study

E.1. Project 24I003 – Internal Order 86988

General description. A transient run-up has been studied on TPI TF_AX3-D pump with Madyn software.

It works at a rated speed of 5220 rpm. The machine train model includes the electric motor, the gearbox rotors and the flexible couplings. The Electric motor rated speed is 1786 rpm while the gear ratio between driven and driver is equal to 2.926. The gearbox has 2 shafts: the wheel shaft coupled with the motor and the pinion shaft coupled with the pump (*Figure 23*).

A VFD varies the pump rotating speed in a range from 4870 to 5220 rpm and regulates the run-up time.

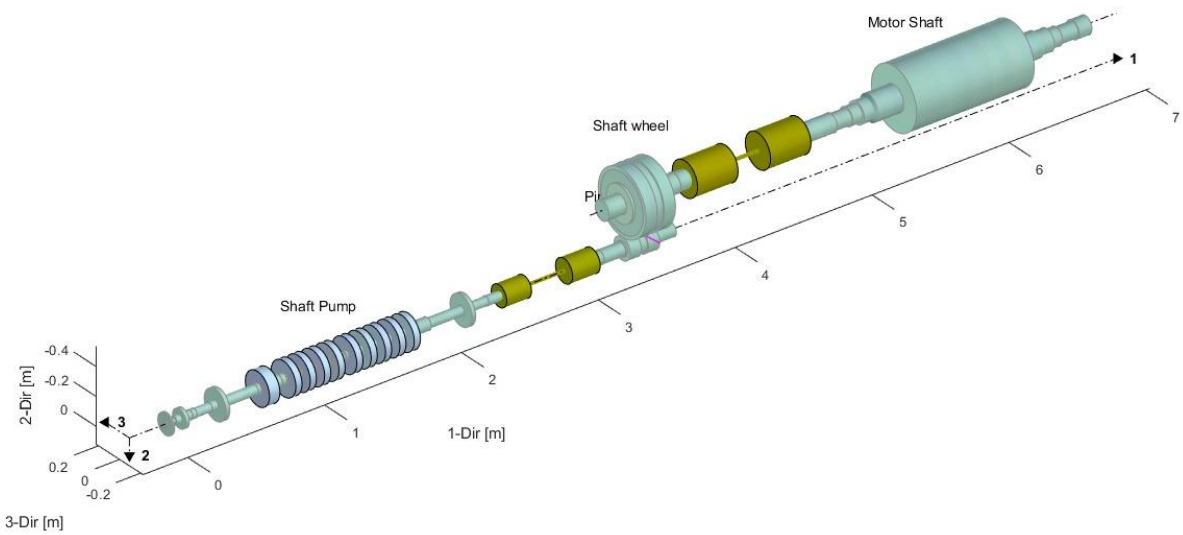

Figure 23 Numerical model

Transient run-up. The driving torque has been defined considering the counter torque shown in *Figure 24*. Oscillating torques multiples of the motor frequency (1X, 2X, 4X, 6X and 12X) and related to VFD with an amplitude of 1% of the rated torque are included in the analysis.

The counter torque normalized ($M_{nominal\ pump} = 2388\ Nm$) in function of speed is defined as follows:

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Driver speed [%]	0	10	20	30	40	50	60	70	80	90	100
$M_{counter}/M_{nominal\ pump}$	0.18	0.09	0.06	0.09	0.16	0.25	0.36	0.49	0.64	0.81	1

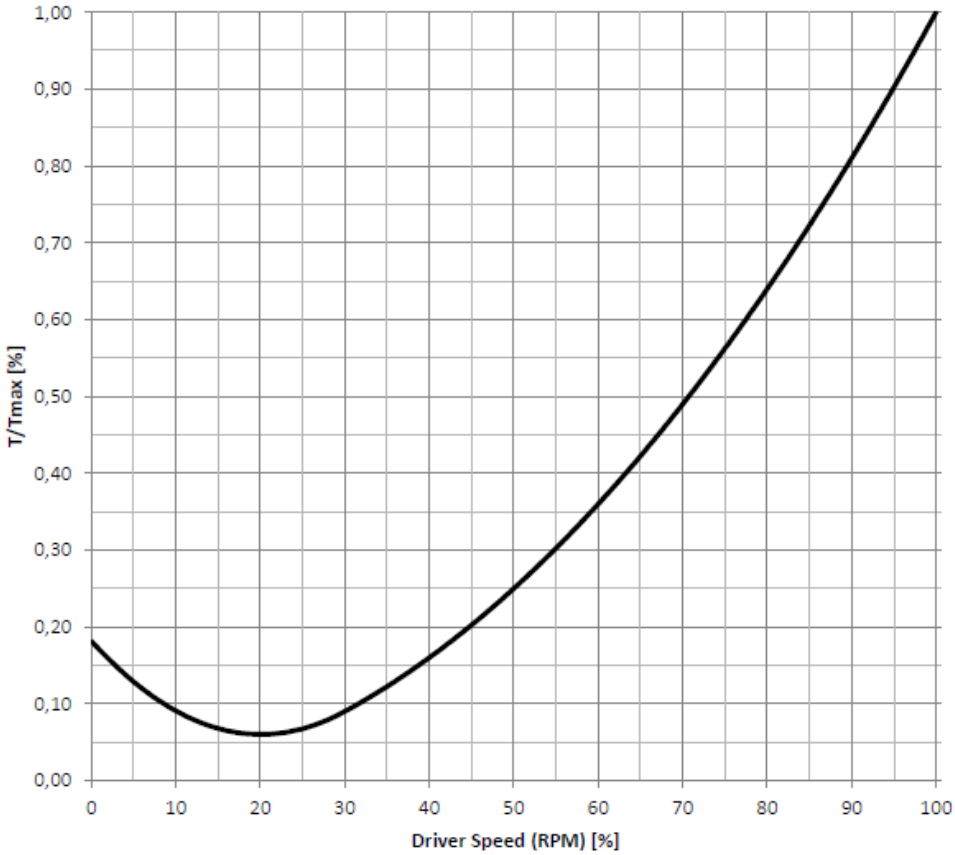

Figure 24. Pump counter torque (open valve)

Two run-up conditions were considered in the analysis:

- a constant driving torque equal to motor nominal torque in all the speed range (#1).
- a parabolic driving torque that follows the pump's curve (#2).

The driving torques normalized ($M_{nominal\ motor} = 7002\ Nm$) in function of the speed are defined as follows:

Driver speed [%]	0	10	20	30	40	50	60	70	80	89.5	100
#1	1	1	1	1	1	1	1	1	1	1	1
#2	0.38	0.29	0.26	0.29	0.36	0.45	0.56	0.69	0.84	1	1

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Figure 25 reports the constant driving torque (blue), the oscillating driving torque (red), the counter torque (green) and the rotor speed (black) in function of time for case #1. A run-up time of 3.7s results from the analysis.

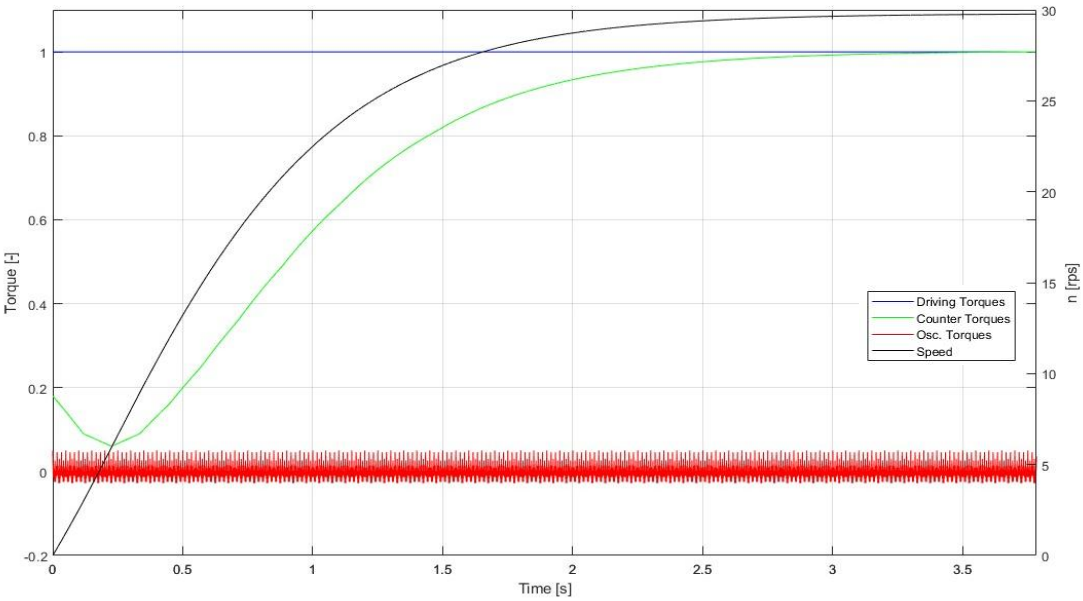

Figure 25. Transient torques normalized #1

Figure 26 shows the trend over time of torques at flexible couplings respectively at the pump's and motor's side. The maximum torque during run-up results equal to 1355.9 Nm on pump side and 6648.3 Nm on motor side.

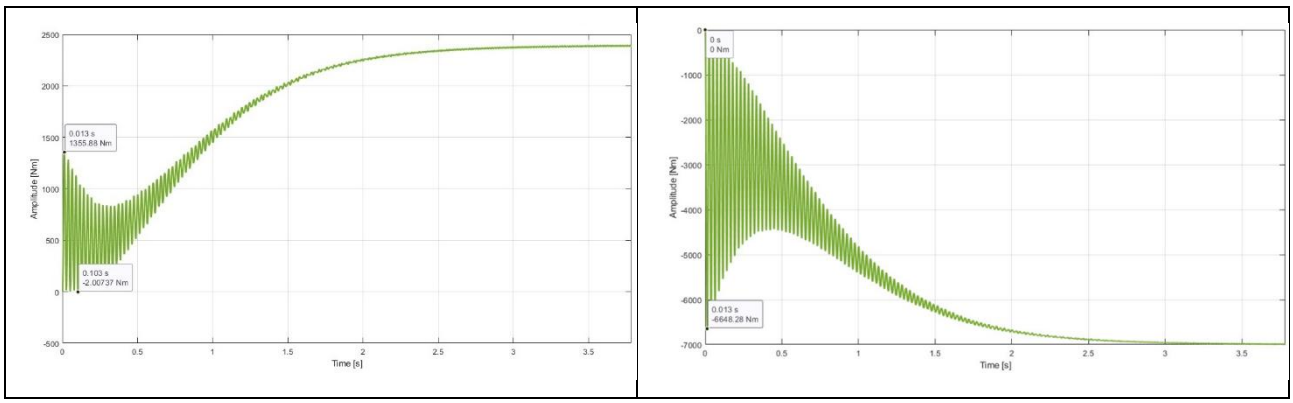

Figure 26. Coupling transient torque #1: pump (left); motor (right)

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The Goodman diagram has been presented in *Figure 27* for a station near the coupling.

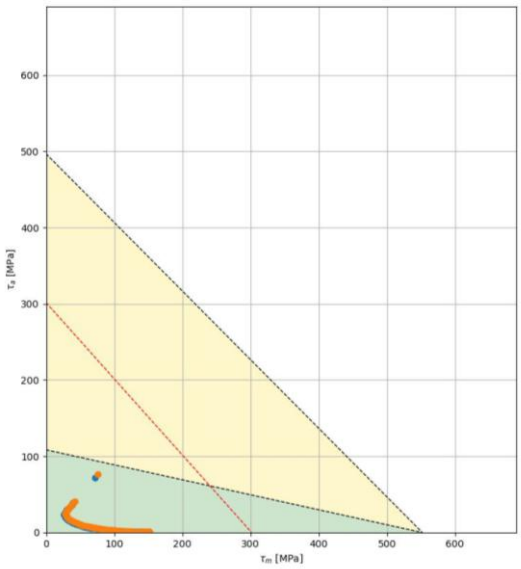

Figure 27. Goodman diagram: run-up #1

Figure 28 reports the driving and counter curves in function of time for case #2. A run-up time of 6.8s results from the analysis.

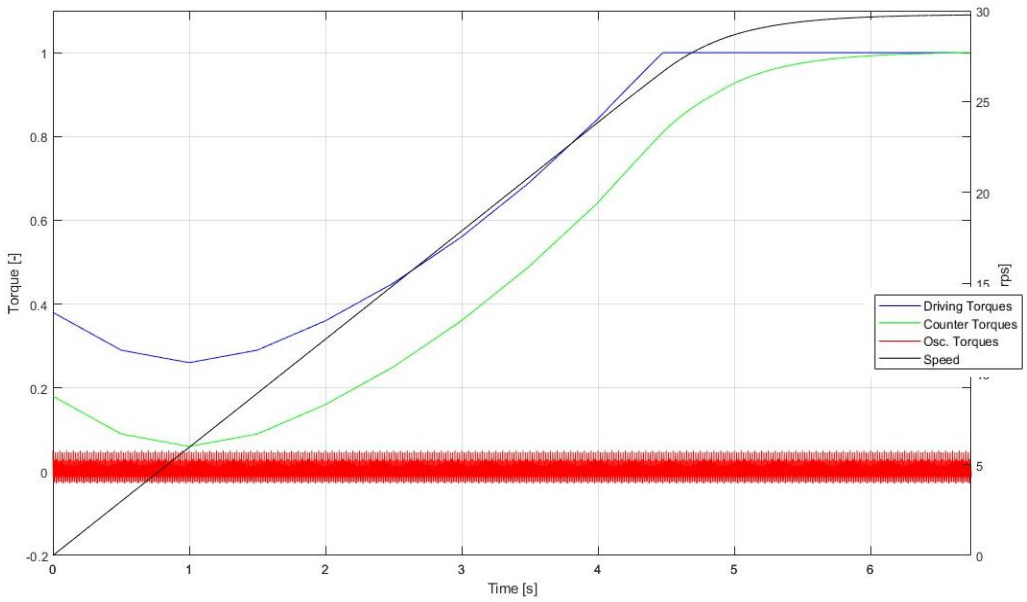

Figure 28. Transient torques normalized #2

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The maximum torque during run-up results equal to 863.0 Nm at the pump's side and 3853.4 Nm at the motor's side.

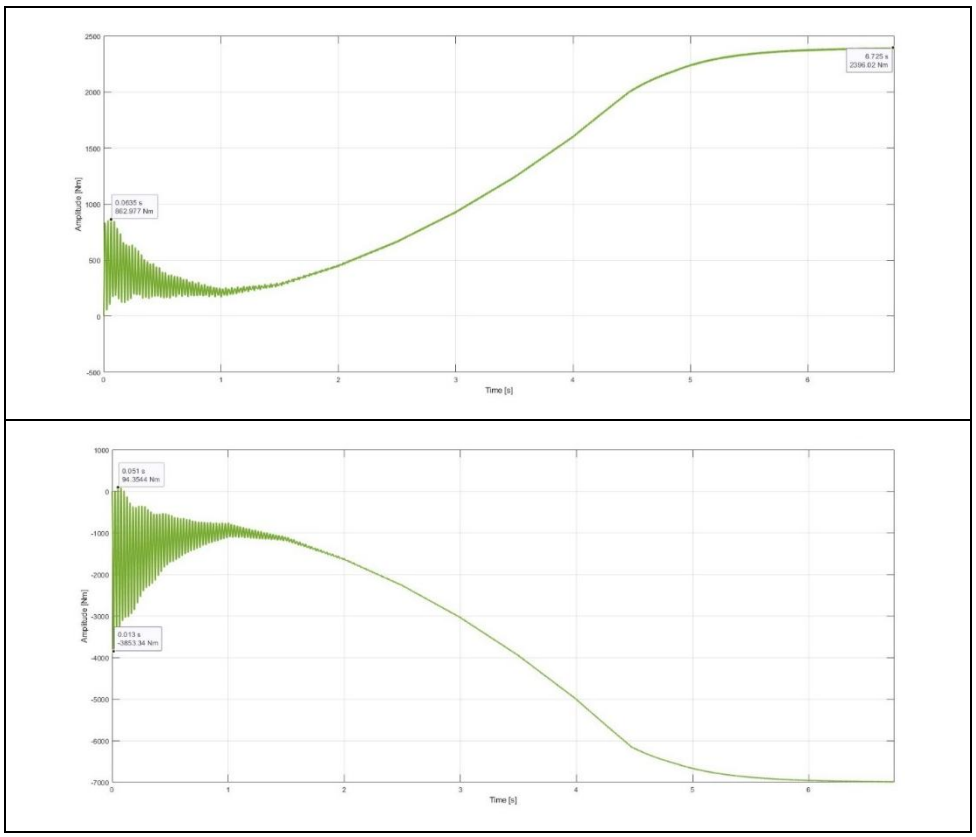

Figure 29. Coupling transient torque #2: pump (left); motor (right)

The Goodman diagram has been presented in *Figure 30* for station 152 near the coupling.

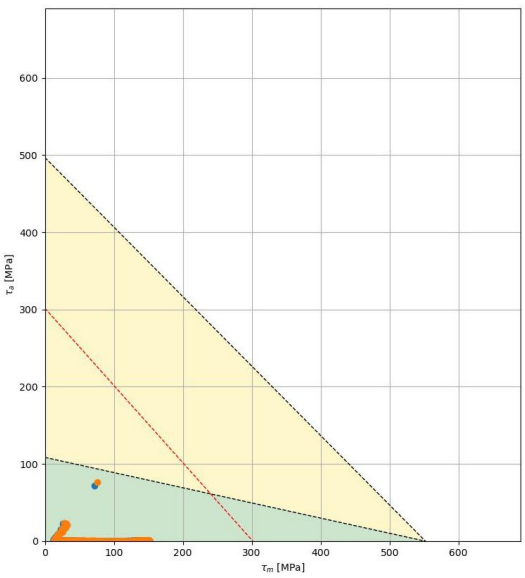

Figure 30. Goodman diagram: run-up #2

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The max peak torque provided by the coupling's supplier is equal to 5508 Nm for the pump's coupling and equal to 34312 Nm for the motor's coupling. The absolute values of the transient torque results are lower than the coupling max peaks provided.

The Goodman diagrams present all points inside the infinite-life region (green zone), meaning the component can support those stress combinations for at least 10^6 cycles. Based on the *Palmgren-Miner's* criteria, the shaft does not present sections with a fatigue life less than 10^6 cycles.

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